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"from sklearn.model_selection import train_test_split, cross_val_score\n",  
"from sklearn.tree import DecisionTreeClassifier\n",  
"from sklearn.ensemble import RandomForestClassifier\n",  
"from xgboost import XGBClassifier\n",  
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```

"    await google.colab.kernel.invokeFunction('convertToInteractive',\n",
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"    '<a target=\"_blank\" href=https://colab.research.google.com/notebooks/data_table.ipynb>data table notebook</a>'\n",
"    + ' to learn more about interactive tables.';\n",
"    element.innerHTML = \";\n",
"    dataTable['output_type'] = 'display_data';\n",
"    await google.colab.output.renderOutput(dataTable, element);\n",
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"     const quickchartButtonEl =\n",
"       document.querySelector('#' + key + ' button');\n",
"     quickchartButtonEl.disabled = true; // To prevent multiple clicks.\n",
"     quickchartButtonEl.classList.add('colab-df-spinner');
```

```

"    try {\n",
"      const charts = await google.colab.kernel.invokeFunction(\n",
"        'suggestCharts', [key], {});\n",
"    } catch (error) {\n",
"      console.error('Error during call to suggestCharts:', error);\n",
"    }\n",
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"    quickchartButtonEl.classList.add('colab-df-quickchart-complete');\n",
"  }\n",
"  (() => {\n",
"    let quickchartButtonEl =\n",
"      document.querySelector('#df-7ef629c8-e1ca-4eb2-be16-bf630391d\nb29 button');\n",
"    quickchartButtonEl.style.display =\n",
"      google.colab.kernel.accessAllowed ? 'block' : 'none';\n",
"  })();\n",
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          "0  7590-VHVEG  Female           0   Yes      No        1        No \\n",
          "1  5575-GNVDE   Male           0   No      No       34        Yes \\n",

```

```

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" MultipleLines InternetService OnlineSecurity OnlineBackup \\n",
"0 No phone service DSL No Yes \n",
"1 No DSL Yes No \n",
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"0 No No No No Month-to-month \n",
"1 Yes No No No One year \n",
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" <th>Dependents</th>\n",
" <th>tenure</th>\n",
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" <td>5575-GNVDE</td>\n",
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" <td>0</td>\n",
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" <td>No</td>\n",
" <td>34</td>\n",
" <td>Yes</td>\n",
" <td>No</td>\n",
" <td>DSL</td>\n",
" <td>Yes</td>\n",
" <td>No</td>\n",
" <td>Yes</td>\n",

```

"    <td>No</td>\n",
"    <td>No</td>\n",
"    <td>No</td>\n",
"    <td>One year</td>\n",
"    <td>No</td>\n",
"    <td>Mailed check</td>\n",
"    <td>56.95</td>\n",
"    <td>1889.5</td>\n",
"    <td>No</td>\n",
"  </tr>\n",
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"</table>\n",
"</div>\n",
"  <div class=\"colab-df-buttons\">\n",
"\n",
"  <div class=\"colab-df-container\">\n",
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"      title=\"Convert this dataframe to an interactive table.\">\n",
"      style=\"display:none;\">\n",
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60v-160H180v160Zm440 0h160v-160H620v160ZM180-180h160v-160H180v16
0Zm440 0h160v-160H620v160Z\"/>\n",
"    </svg>\n",
"  </button>\n",
"\n",
"  <style>\n",
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"      display:flex;\n",

```

```
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```

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"   fill: #FFFFFF;\n",
" }\n",
" </style>\n",
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"     document.querySelector('#df-ca357cb3-ad61-4ac8-8b57-f1f14eede5af button.colab-df-convert');\n",
"   buttonEl.style.display =\n",
"     google.colab.kernel.accessAllowed ? 'block' : 'none';\n",
"\n",
"   async function convertToInteractive(key) {\n",
"     const element = document.querySelector('#df-ca357cb3-ad61-4ac8-8b57-f1f14eede5af');\n",
"     const dataTable =\n",
"       await google.colab.kernel.invokeFunction('convertToInteractive',\n",
"         [key], {});\n",
"     if (!dataTable) return;\n",
"\n",
"     const docLinkHtml = 'Like what you see? Visit the ' +\n",
"       '<a target=\"_blank\" href=https://colab.research.google.com/notebooks/data_table.ipynb>data table notebook</a>'\n",
"       + ' to learn more about interactive tables.';\n",
"     element.innerHTML = \"\n",
"       dataTable['output_type'] = 'display_data';\n",
"       await google.colab.output.renderOutput(dataTable, element);\n",
"       const docLink = document.createElement('div');\n",
"       docLink.innerHTML = docLinkHtml;\n",
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"     }\n",
" </script>\n",

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"    --hover-fill-color: #174EA6;\n",
"    --disabled-fill-color: #AAA;\n",
"    --disabled-bg-color: #DDD;\n",
"  }\n",
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"   display: none;\n",  
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" }\n",  
"\n",  
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"   fill: var(--disabled-fill-color);\n",  
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" }\n",  
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"   0% {\n",  
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"   }\n",  
"   20% {\n",  
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"     border-top-color: var(--fill-color);\n",  
"   }\n",  
"   30% {\n",  
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"     border-top-color: var(--fill-color);\n",  
"     border-right-color: var(--fill-color);\n",  
"   }\n",  
"   40% {\n",  
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"     border-right-color: var(--fill-color);\n",  
"     border-top-color: var(--fill-color);\n",  
"   }\n",  
"   60% {\n",  
"     border-color: transparent;\n",  
"     border-right-color: var(--fill-color);\n",  
"   }\n",  
"   80% {\n",
```

```

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" border-bottom-color: var(--fill-color);\n",
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"\n",
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" async function quickchart(key) {\n",
" const quickchartButtonEl =\n",
" document.querySelector('#' + key + ' button');\n",
" quickchartButtonEl.disabled = true; // To prevent multiple clicks.\n",
" quickchartButtonEl.classList.add('colab-df-spinner');\n",
" try {\n",
" const charts = await google.colab.kernel.invokeFunction(\n",
" 'suggestCharts', [key], {});\n",
" } catch (error) {\n",
" console.error('Error during call to suggestCharts:', error);\n",
" }\n",
" quickchartButtonEl.classList.remove('colab-df-spinner');\n",
" quickchartButtonEl.classList.add('colab-df-quickchart-complete');\n",
" }\n",
" (() => {\n",
" let quickchartButtonEl =\n",
" document.querySelector('#df-840b0645-bdab-405e-ab5d-e3bfaf2\n526d3 button');\n",
" quickchartButtonEl.style.display =\n",
" google.colab.kernel.accessAllowed ? 'block' : 'none';\n",
" })();\n",

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  1   gender          7043 non-null  object \n",
  2   SeniorCitizen   7043 non-null  int64  \n",
  3   Partner         7043 non-null  object \n",
  4   Dependents      7043 non-null  object \n",
  5   tenure          7043 non-null  int64  \n",
  6   PhoneService    7043 non-null  object \n",
  7   MultipleLines   7043 non-null  object \n",
  8   InternetService 7043 non-null  object \n",
  9   OnlineSecurity  7043 non-null  object \n",
  10  OnlineBackup    7043 non-null  object \n",
  11  DeviceProtection 7043 non-null  object \n",
  12  TechSupport     7043 non-null  object \n",
  13  StreamingTV     7043 non-null  object \n",
  14  StreamingMovies 7043 non-null  object \n",
  15  Contract        7043 non-null  object \n",
  16  PaperlessBilling 7043 non-null  object \n",
  17  PaymentMethod   7043 non-null  object \n",
  18  MonthlyCharges  7043 non-null  float64\n",
  19  TotalCharges    7043 non-null  object \n",
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]
}
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},

```

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```

```

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"0 No phone service      DSL      No      Yes \n",
"1      No      DSL      Yes      No \n",
"\n",
"  DeviceProtection TechSupport StreamingTV StreamingMovies  Contr
act \\n",
"0      No      No      No      No Month-to-month \n",
"1      Yes      No      No      No  One year \n",
"\n",
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```

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"      <th>Partner</th>\n",
"      <th>Dependents</th>\n",
"      <th>tenure</th>\n",
"      <th>PhoneService</th>\n",
"      <th>MultipleLines</th>\n",
"      <th>InternetService</th>\n",
"      <th>OnlineSecurity</th>\n",
"      <th>OnlineBackup</th>\n",
"      <th>DeviceProtection</th>\n",
"      <th>TechSupport</th>\n",
"      <th>StreamingTV</th>\n",
"      <th>StreamingMovies</th>\n",
"      <th>Contract</th>\n",
"      <th>PaperlessBilling</th>\n",
"      <th>PaymentMethod</th>\n",
"      <th>MonthlyCharges</th>\n",
"      <th>TotalCharges</th>\n",
"      <th>Churn</th>\n",
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```


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" <td>No</td>\n",
" <td>No</td>\n",
" <td>No</td>\n",
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```

"    <td>No</td>\n",
"    <td>One year</td>\n",
"    <td>No</td>\n",
"    <td>Mailed check</td>\n",
"    <td>56.95</td>\n",
"    <td>1889.5</td>\n",
"    <td>No</td>\n",
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"    fill: #174EA6;\n",
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```

```
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"      const element = document.querySelector('#df-0f036ad4-f974-4e02-\n93d3-f468fc4954b6');\n",
"      const dataTable =\n",
"        await google.colab.kernel.invokeFunction('convertToInteractive',\n",
"          [key], {});\n",
"      if (!dataTable) return;\n",
"\n",
"      const docLinkHtml = 'Like what you see? Visit the ' +\n",
"        '<a target=\"_blank\" href=https://colab.research.google.com/noteb\nooks/data_table.ipynb>data table notebook</a>'\n",
"        + ' to learn more about interactive tables.';\n",
"      element.innerHTML =\n",
"        dataTable['output_type'] = 'display_data';\n",
"      await google.colab.output.renderOutput(dataTable, element);\n",
"      const docLink = document.createElement('div');\n",
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"  height: 32px;\n",
"  padding: 0;\n",
"  width: 32px;\n",
"}\n",
"\n",
".colab-df-quickchart:hover {\n",
"  background-color: var(--hover-bg-color);\n",
"  box-shadow: 0 1px 2px rgba(60, 64, 67, 0.3), 0 1px 3px 1px rgba(60, 64, 67, 0.15);\n",
"  fill: var(--button-hover-fill-color);\n",
"}\n",
"\n",
".colab-df-quickchart-complete:disabled,\n",
".colab-df-quickchart-complete:disabled:hover {\n",
"  background-color: var(--disabled-bg-color);\n",
"  fill: var(--disabled-fill-color);\n",
"  box-shadow: none;\n",
"}\n",
"\n",
".colab-df-spinner {\n",
"  border: 2px solid var(--fill-color);\n",
"  border-color: transparent;\n",
"  border-bottom-color: var(--fill-color);
```

```
" animation:\n",  
"   spin 1s steps(1) infinite;\n",  
" }\n",  
"\n",  
" @keyframes spin {\n",  
"   0% {\n",  
"     border-color: transparent;\n",  
"     border-bottom-color: var(--fill-color);\n",  
"     border-left-color: var(--fill-color);\n",  
"   }\n",  
"   20% {\n",  
"     border-color: transparent;\n",  
"     border-left-color: var(--fill-color);\n",  
"     border-top-color: var(--fill-color);\n",  
"   }\n",  
"   30% {\n",  
"     border-color: transparent;\n",  
"     border-left-color: var(--fill-color);\n",  
"     border-top-color: var(--fill-color);\n",  
"     border-right-color: var(--fill-color);\n",  
"   }\n",  
"   40% {\n",  
"     border-color: transparent;\n",  
"     border-right-color: var(--fill-color);\n",  
"     border-top-color: var(--fill-color);\n",  
"   }\n",  
"   60% {\n",  
"     border-color: transparent;\n",  
"     border-right-color: var(--fill-color);\n",  
"   }\n",  
"   80% {\n",  
"     border-color: transparent;\n",  
"     border-right-color: var(--fill-color);
```

```

" border-bottom-color: var(--fill-color);\n",
" }\n",
" 90% {\n",
" border-color: transparent;\n",
" border-bottom-color: var(--fill-color);\n",
" }\n",
" }\n",
"</style>\n",
"\n",
" <script>\n",
" async function quickchart(key) {\n",
" const quickchartButtonEl =\n",
" document.querySelector('#' + key + ' button');\n",
" quickchartButtonEl.disabled = true; // To prevent multiple clicks.\n",
" quickchartButtonEl.classList.add('colab-df-spinner');\n",
" try {\n",
" const charts = await google.colab.kernel.invokeFunction(\n",
" 'suggestCharts', [key], {});\n",
" } catch (error) {\n",
" console.error('Error during call to suggestCharts:', error);\n",
" }\n",
" quickchartButtonEl.classList.remove('colab-df-spinner');\n",
" quickchartButtonEl.classList.add('colab-df-quickchart-complete');\n",
" }\n",
" (() => {\n",
" let quickchartButtonEl =\n",
" document.querySelector('#df-2834a9c3-1df7-4537-b6f3-b2e5aa2d4355 button');\n",
" quickchartButtonEl.style.display =\n",
" google.colab.kernel.accessAllowed ? 'block' : 'none';\n",
" })();\n",
" </script>\n",
"</div>\n",

```



```

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```

```
{\n  \"column\": \"OnlineSecurity\", \n  \"properties\": {\n    \"dtype\": \"category\", \n    \"num_unique_values\": 3, \n    \"samples\": [\n      \"No\", \n      \"Yes\" \n    ], \n    \"semantic_type\": \"\", \n    \"description\": \"\" \n  } \n}, \n {\n  \"column\": \"OnlineBackup\", \n  \"properties\": {\n    \"dtype\": \"category\", \n    \"num_unique_values\": 3, \n    \"samples\": [\n      \"Yes\", \n      \"No\" \n    ], \n    \"semantic_type\": \"\", \n    \"description\": \"\" \n  } \n}, \n {\n  \"column\": \"DeviceProtection\", \n  \"properties\": {\n    \"dtype\": \"category\", \n    \"num_unique_values\": 3, \n    \"samples\": [\n      \"No\", \n      \"Yes\" \n    ], \n    \"semantic_type\": \"\", \n    \"description\": \"\" \n  } \n}, \n {\n  \"column\": \"TechSupport\", \n  \"properties\": {\n    \"dtype\": \"category\", \n    \"num_unique_values\": 3, \n    \"samples\": [\n      \"No\", \n      \"Yes\" \n    ], \n    \"semantic_type\": \"\", \n    \"description\": \"\" \n  } \n}, \n {\n  \"column\": \"StreamingTV\", \n  \"properties\": {\n    \"dtype\": \"category\", \n    \"num_unique_values\": 3, \n    \"samples\": [\n      \"No\", \n      \"Yes\" \n    ], \n    \"semantic_type\": \"\", \n    \"description\": \"\" \n  } \n}, \n {\n  \"column\": \"StreamingMovies\", \n  \"properties\": {\n    \"dtype\": \"category\", \n    \"num_unique_values\": 3, \n    \"samples\": [\n      \"No\", \n      \"Yes\" \n    ], \n    \"semantic_type\": \"\", \n    \"description\": \"\" \n  } \n}, \n {\n  \"column\": \"Contract\", \n  \"properties\": {\n    \"dtype\": \"category\", \n    \"num_unique_values\": 3, \n    \"samples\": [\n      \"Month-to-month\", \n      \"One year\" \n    ], \n    \"semantic_type\": \"\", \n    \"description\": \"\" \n  } \n}, \n {\n  \"column\": \"PaperlessBilling\", \n  \"properties\": {\n    \"dtype\": \"category\", \n    \"num_unique_values\": 2, \n    \"samples\": [\n      \"No\", \n      \"Yes\" \n    ], \n    \"semantic_type\": \"\", \n    \"description\": \"\" \n  } \n}, \n {\n  \"column\": \"PaymentMethod\", \n  \"properties\": {\n    \"dtype\": \"category\", \n    \"num_unique_values\": 4, \n    \"samples\": [\n      \"Mailed check\", \n      \"Credit card (automatic)\" \n    ], \n    \"semantic_type\": \"\", \n    \"description\": \"\" \n  } \n}, \n {\n  \"column\": \"MonthlyCharges\", \n  \"properties\": {\n    \"dtype\": \"number\", \n    \"std\": 30.09004709767854, \n    \"min\": 18.25, \n    \"max\": 118.75, \n    \"num_unique_values\": 1585, \n    \"samples\": [\n      48.85, \n      20.05 \n    ], \n    \"semantic_type\": \"\", \n    \"description\": \"\" \n  } \n}, \n {\n  \"column\": \"TotalCharges\", \n  \"properties\": {\n    \"dtype\": \"string\", \n    \"num_unique_values\": 6531, \n    \"sa
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```

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```

```

        "    'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod',\n",
        "    'MonthlyCharges', 'TotalCharges', 'Churn'],\n",
        "    dtype='object')"
```

]

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"execution_count": 10

}

]

},

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"print(df[\"gender\"].unique())"

],

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},

"execution_count": 11,

"outputs": [

{

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"name": "stdout",

"text": [

"['Female' 'Male']\n"

]

}

]

},

```

{
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  ],
  "metadata": {
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    },
    "id": "IKiFNy2rLEUx",
    "outputId": "c420d9a6-0d4b-4660-986a-f9e28f7a877b"
  },
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      "name": "stdout",
      "text": [
        "[0 1]\n"
      ]
    }
  ],
},
{
  "cell_type": "code",
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    "# printing the unique values in all the columns\n",
    "\n",
    "numerical_features_list = [\"tenure\", \"MonthlyCharges\", \"TotalCharges\"]\n",
    "\n",
    "for col in df.columns:\n",
    "    if col not in numerical_features_list:\n",

```

```
" print(col, df[col].unique())\n",
" print(\"-\"*50)"
],
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  "id": "heeFR4JJKULo",
  "outputId": "53195360-e58a-4c92-b212-b030ba51cf2c"
},
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"outputs": [
  {
    "output_type": "stream",
    "name": "stdout",
    "text": [
      "gender ['Female' 'Male']\n",
      "-----\n",
      "SeniorCitizen [0 1]\n",
      "-----\n",
      "Partner ['Yes' 'No']\n",
      "-----\n",
      "Dependents ['No' 'Yes']\n",
      "-----\n",
      "PhoneService ['No' 'Yes']\n",
      "-----\n",
      "MultipleLines ['No phone service' 'No' 'Yes']\n",
      "-----\n",
      "InternetService ['DSL' 'Fiber optic' 'No']\n",
      "-----\n",
      "OnlineSecurity ['No' 'Yes' 'No internet service']\n",
      "-----\n",
      "OnlineBackup ['Yes' 'No' 'No internet service']\n",
```

```

"-----\n",
"DeviceProtection ['No' 'Yes' 'No internet service']\n",
"-----\n",
"TechSupport ['No' 'Yes' 'No internet service']\n",
"-----\n",
"StreamingTV ['No' 'Yes' 'No internet service']\n",
"-----\n",
"StreamingMovies ['No' 'Yes' 'No internet service']\n",
"-----\n",
"Contract ['Month-to-month' 'One year' 'Two year']\n",
"-----\n",
"PaperlessBilling ['Yes' 'No']\n",
"-----\n",
"PaymentMethod ['Electronic check' 'Mailed check' 'Bank transfer (automatic)'\n",
" 'Credit card (automatic)']\n",
"-----\n",
"Churn ['No' 'Yes']\n",
"-----\n"
]
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      "SeniorCitizen  0\n",
      "Partner      0\n",
      "Dependents    0\n",
      "tenure      0\n",
      "PhoneService  0\n",
      "MultipleLines 0\n",
      "InternetService 0\n",
      "OnlineSecurity 0\n",
      "OnlineBackup  0\n",
      "DeviceProtection 0\n",
      "TechSupport   0\n",
      "StreamingTV   0\n",
      "StreamingMovies 0\n",
      "Contract      0\n",
      "PaperlessBilling 0\n",
      "PaymentMethod 0\n",
      "MonthlyCharges 0\n",
      "TotalCharges  0\n",
      "Churn         0\n",
      "dtype: int64\n"
    ]
  }
],
},
```



```

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  ],
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  "outputs": []
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    "df[df[\"TotalCharges\"]==\" \"]"
  ],
  "metadata": {
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      "height": 570
    },
    "id": "gvHFhTBRMJJt",
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      "output_type": "execute_result",
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          "   gender SeniorCitizen Partner Dependents tenure PhoneService \\n",
          "488  Female           0   Yes     Yes     0       No  \n",
          "753  Male           0   No     Yes     0       Yes  \n",

```

"936 Female 0 Yes Yes 0 Yes \n",
 "1082 Male 0 Yes Yes 0 Yes \n",
 "1340 Female 0 Yes Yes 0 No \n",
 "3331 Male 0 Yes Yes 0 Yes \n",
 "3826 Male 0 Yes Yes 0 Yes \n",
 "4380 Female 0 Yes Yes 0 Yes \n",
 "5218 Male 0 Yes Yes 0 Yes \n",
 "6670 Female 0 Yes Yes 0 Yes \n",
 "6754 Male 0 No Yes 0 Yes \n",
 "\n",

" MultipleLines InternetService OnlineSecurity \\n",
 "488 No phone service DSL Yes \n",
 "753 No No No internet service \n",
 "936 No DSL Yes \n",
 "1082 Yes No No internet service \n",
 "1340 No phone service DSL Yes \n",
 "3331 No No No internet service \n",
 "3826 Yes No No internet service \n",
 "4380 No No No internet service \n",
 "5218 No No No internet service \n",
 "6670 Yes DSL No \n",
 "6754 Yes DSL Yes \n",
 "\n",

" OnlineBackup DeviceProtection TechSupport \\n",
 "488 No Yes Yes \n",
 "753 No internet service No internet service No internet service \n",
 "936 Yes Yes No \n",
 "1082 No internet service No internet service No internet service \n",
 "1340 Yes Yes Yes \n",
 "3331 No internet service No internet service No internet service \n",
 "3826 No internet service No internet service No internet service \n",
 "4380 No internet service No internet service No internet service \n",
 "5218 No internet service No internet service No internet service \n",

```

"6670      Yes      Yes      Yes \n",
"6754      Yes      No      Yes \n",
"\n",
"      StreamingTV      StreamingMovies      Contract PaperlessBilling \\\n",
"488      Yes      No Two year      Yes \n",
"753      No internet service      No internet service      Two year      No \n",
"936      Yes      Yes Two year      No \n",
"1082      No internet service      No internet service      Two year      No \n",
"1340      Yes      No Two year      No \n",
"3331      No internet service      No internet service      Two year      No \n",
"3826      No internet service      No internet service      Two year      No \n",
"4380      No internet service      No internet service      Two year      No \n",
"5218      No internet service      No internet service      One year      Yes \n",
"6670      Yes      No Two year      No \n",
"6754      No      No Two year      Yes \n",
"\n",

```

```

"      PaymentMethod      MonthlyCharges      TotalCharges      Churn \n",
"488      Bank transfer (automatic)      52.55      No \n",
"753      Mailed check      20.25      No \n",
"936      Mailed check      80.85      No \n",
"1082      Mailed check      25.75      No \n",
"1340      Credit card (automatic)      56.05      No \n",
"3331      Mailed check      19.85      No \n",
"3826      Mailed check      25.35      No \n",
"4380      Mailed check      20.00      No \n",
"5218      Mailed check      19.70      No \n",
"6670      Mailed check      73.35      No \n",
"6754      Bank transfer (automatic)      61.90      No "

```

```

],

```

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"     vertical-align: middle;\n",
" } \n",
"\n",
" .dataframe tbody tr th {\n",
"     vertical-align: top;\n",
" } \n",
"\n",
" .dataframe thead th {\n",
"     text-align: right;\n",
" } \n",
"</style>\n",
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"     <th>SeniorCitizen</th>\n",
"     <th>Partner</th>\n",
"     <th>Dependents</th>\n",
"     <th>tenure</th>\n",
"     <th>PhoneService</th>\n",
"     <th>MultipleLines</th>\n",
"     <th>InternetService</th>\n",
"     <th>OnlineSecurity</th>\n",
"     <th>OnlineBackup</th>\n",
"     <th>DeviceProtection</th>\n",
"     <th>TechSupport</th>\n",
"     <th>StreamingTV</th>\n",
"     <th>StreamingMovies</th>\n",
"     <th>Contract</th>
```

```

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" <th>TotalCharges</th>\n",
" <th>Churn</th>\n",
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" <td>Yes</td>\n",
" <td>Yes</td>\n",
" <td>Yes</td>\n",
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" <td>Two year</td>\n",
" <td>Yes</td>\n",
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" <tr>\n",
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```

```
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" <td>No</td>\n",
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" <td>Yes</td>\n",
" <td>Yes</td>\n",
```

" <td>Yes</td>\n",
" <td>No</td>\n",
" <td>Yes</td>\n",
" <td>Yes</td>\n",
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" <td>No internet service</td>\n",
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"   <td>0</td>\n",
"   <td>No</td>\n",
"   <td>No phone service</td>\n",
"   <td>DSL</td>\n",
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"   <td>Yes</td>\n",
"   <td>Yes</td>\n",
"   <td>Yes</td>\n",
"   <td>Yes</td>\n",
"   <td>No</td>\n",
"   <td>Two year</td>\n",
"   <td>No</td>\n",
"   <td>Credit card (automatic)</td>\n",
"   <td>56.05</td>\n",
"   <td></td>\n",
"   <td>No</td>\n",
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"   <td>Yes</td>\n",
"   <td>0</td>\n",
"   <td>Yes</td>\n",
"   <td>No</td>
```


" <td>No</td>\n",
" <td>No internet service</td>\n",
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" <td>No internet service</td>\n",
" <td>No internet service</td>\n",
" <td>No internet service</td>\n",
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" <td>Yes</td>\n",
" <td>Yes</td>\n",
" <td>No</td>\n",
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" <td>No internet service</td>\n",
" <td>No internet service</td>\n",
" <td>No internet service</td>\n",
" <td>No internet service</td>\n",
" <td>Two year</td>\n",
" <td>No</td>\n",
" <td>Mailed check</td>\n",

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"    <td>No</td>\n",
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" <tr>\n",
"    <th>4380</th>\n",
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"    <td>No internet service</td>\n",
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"    <td>Yes</td>
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" <td>Yes</td>\n",
" <td>Yes</td>\n",
" <td>Yes</td>\n",
" <td>No</td>\n",
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```
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"   <td>No</td>\n",
"   <td>Yes</td>\n",
"   <td>No</td>\n",
"   <td>No</td>\n",
"   <td>Two year</td>\n",
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"   <td>Bank transfer (automatic)</td>\n",
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" </tbody>\n",
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" <div class=\"colab-df-buttons\">\n",
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" <div class=\"colab-df-container\">\n",
" <button class=\"colab-df-convert\" onclick=\"convertToInteractive('df-3
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"     title=\"Convert this dataframe to an interactive table.\" \n",
"     style=\"display:none;\">\n",
"\n",
" <svg xmlns=\"http://www.w3.org/2000/svg\" height=\"24px\" viewBox=
\"0 960 960 960\">\n",
" <path d=\"M120-120v-720h720v720H120Zm60-500h600v-160H180v1
60Zm220 220h160v-160H400v160Zm0 220h160v-160H400v160ZM180-400h1
60v-160H180v160Zm440 0h160v-160H620v160ZM180-180h160v-160H180v16
0Zm440 0h160v-160H620v160Z\"/>\n",
" </svg>\n",
" </button>\n",
"\n",
" <style>\n",
" .colab-df-container {\n",
"   display: flex;\n",
"   gap: 12px;\n",
" }\n",
"\n",
" .colab-df-convert {\n",
"   background-color: #E8F0FE;\n",
"   border: none;\n",
"   border-radius: 50%;\n",
"   cursor: pointer;\n",
"   display: none;\n",
"   fill: #1967D2;\n",
"   height: 32px;\n",
"   padding: 0 0 0 0;\n",
"   width: 32px;\n",
"
```

```
" }\n",
"\n",
" .colab-df-convert:hover {\n",
"   background-color: #E2EBFA;\n",
"   box-shadow: 0px 1px 2px rgba(60, 64, 67, 0.3), 0px 1px 3px 1px rgba(60, 64, 67, 0.15);\n",
"   fill: #174EA6;\n",
" }\n",
"\n",
" .colab-df-buttons div {\n",
"   margin-bottom: 4px;\n",
" }\n",
"\n",
" [theme=dark] .colab-df-convert {\n",
"   background-color: #3B4455;\n",
"   fill: #D2E3FC;\n",
" }\n",
"\n",
" [theme=dark] .colab-df-convert:hover {\n",
"   background-color: #434B5C;\n",
"   box-shadow: 0px 1px 3px 1px rgba(0, 0, 0, 0.15);\n",
"   filter: drop-shadow(0px 1px 2px rgba(0, 0, 0, 0.3));\n",
"   fill: #FFFFFF;\n",
" }\n",
" </style>\n",
"\n",
" <script>\n",
"   const buttonEl =\n",
"     document.querySelector('#df-38c24c03-e456-420d-9cf2-446461b3fb35 button.colab-df-convert');\n",
"   buttonEl.style.display =\n",
"     google.colab.kernel.accessAllowed ? 'block' : 'none';\n",
"
```

```

"    async function convertToInteractive(key) {\n",
"        const element = document.querySelector('#df-38c24c03-e456-420d-9cf2-446461b3fb35');\n",
"        const dataTable =\n",
"            await google.colab.kernel.invokeFunction('convertToInteractive',\n",
"                [key], {});\n",
"        if (!dataTable) return;\n",
"\n",
"        const docLinkHtml = 'Like what you see? Visit the ' +\n",
"            '<a target=\"_blank\" href=https://colab.research.google.com/notebooks/data_table.ipynb>data table notebook</a>'\n",
"            + ' to learn more about interactive tables.';\n",
"        element.innerHTML = \";\n",
"        dataTable['output_type'] = 'display_data';\n",
"        await google.colab.output.renderOutput(dataTable, element);\n",
"        const docLink = document.createElement('div');\n",
"        docLink.innerHTML = docLinkHtml;\n",
"        element.appendChild(docLink);\n",
"    }\n",
"  </script>\n",
" </div>\n",
"\n",
"\n",
"<div id=\"df-228142a6-1a4b-45e4-91a4-24af49d57d2e\">\n",
"  <button class=\"colab-df-quickchart\" onclick=\"quickchart('df-228142a6-1a4b-45e4-91a4-24af49d57d2e')\" \n",
"    title=\"Suggest charts\" \n",
"    style=\"display:none\">\n",
"\n",
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"  width=\"24px\">\n",
"  <g>\n",

```

```
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5c0-1.1-.9-2-2-2zM9 17H7v-7h2v7zm4 0h-2V7h2v10zm4 0h-2v-4h2v4z\"/>\n",
"  </g>\n",
"</svg>\n",
" </button>\n",
"\n",
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"  .colab-df-quickchart {\n",
"    --bg-color: #E8F0FE;\n",
"    --fill-color: #1967D2;\n",
"    --hover-bg-color: #E2EBFA;\n",
"    --hover-fill-color: #174EA6;\n",
"    --disabled-fill-color: #AAA;\n",
"    --disabled-bg-color: #DDD;\n",
"  }\n",
"\n",
"  [theme=dark] .colab-df-quickchart {\n",
"    --bg-color: #3B4455;\n",
"    --fill-color: #D2E3FC;\n",
"    --hover-bg-color: #434B5C;\n",
"    --hover-fill-color: #FFFFFF;\n",
"    --disabled-bg-color: #3B4455;\n",
"    --disabled-fill-color: #666;\n",
"  }\n",
"\n",
"  .colab-df-quickchart {\n",
"    background-color: var(--bg-color);\n",
"    border: none;\n",
"    border-radius: 50%;\n",
"    cursor: pointer;\n",
"    display: none;\n",
"    fill: var(--fill-color);\n",
"    height: 32px;\n",

```



```
" padding: 0;\n",\n" width: 32px;\n",\n" }\n",\n"\n",\n" .colab-df-quickchart:hover {\n",\n" background-color: var(--hover-bg-color);\n",\n" box-shadow: 0 1px 2px rgba(60, 64, 67, 0.3), 0 1px 3px 1px rgba(60, 64,\n67, 0.15);\n",\n" fill: var(--button-hover-fill-color);\n",\n" }\n",\n"\n",\n" .colab-df-quickchart-complete:disabled,\n",\n" .colab-df-quickchart-complete:disabled:hover {\n",\n" background-color: var(--disabled-bg-color);\n",\n" fill: var(--disabled-fill-color);\n",\n" box-shadow: none;\n",\n" }\n",\n"\n",\n" .colab-df-spinner {\n",\n" border: 2px solid var(--fill-color);\n",\n" border-color: transparent;\n",\n" border-bottom-color: var(--fill-color);\n",\n" animation:\n",\n" spin 1s steps(1) infinite;\n",\n" }\n",\n"\n",\n" @keyframes spin {\n",\n" 0% {\n",\n" border-color: transparent;\n",\n" border-bottom-color: var(--fill-color);\n",\n" border-left-color: var(--fill-color);\n",\n" }\n",\n" 20% {\n",\n"
```

```
" border-color: transparent;\n",
" border-left-color: var(--fill-color);\n",
" border-top-color: var(--fill-color);\n",
" }\n",
" 30% {\n",
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" border-left-color: var(--fill-color);\n",
" border-top-color: var(--fill-color);\n",
" border-right-color: var(--fill-color);\n",
" }\n",
" 40% {\n",
" border-color: transparent;\n",
" border-right-color: var(--fill-color);\n",
" border-top-color: var(--fill-color);\n",
" }\n",
" 60% {\n",
" border-color: transparent;\n",
" border-right-color: var(--fill-color);\n",
" }\n",
" 80% {\n",
" border-color: transparent;\n",
" border-right-color: var(--fill-color);\n",
" border-bottom-color: var(--fill-color);\n",
" }\n",
" 90% {\n",
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" border-bottom-color: var(--fill-color);\n",
" }\n",
" }\n",
"</style>\n",
"\n",
" <script>\n",
" async function quickchart(key) {\n",
```

```

"    const quickchartButtonEl =\n",
"    document.querySelector('#' + key + ' button');\n",
"    quickchartButtonEl.disabled = true; // To prevent multiple clicks.\n",
"    quickchartButtonEl.classList.add('colab-df-spinner');\n",
"    try {\n",
"        const charts = await google.colab.kernel.invokeFunction(\n",
"            'suggestCharts', [key], {});\n",
"    } catch (error) {\n",
"        console.error('Error during call to suggestCharts:', error);\n",
"    }\n",
"    quickchartButtonEl.classList.remove('colab-df-spinner');\n",
"    quickchartButtonEl.classList.add('colab-df-quickchart-complete');\n",
"    }\n",
"    (() => {\n",
"        let quickchartButtonEl =\n",
"        document.querySelector('#df-228142a6-1a4b-45e4-91a4-24af49d57d2e button');\n",
"        quickchartButtonEl.style.display =\n",
"        google.colab.kernel.accessAllowed ? 'block' : 'none';\n",
"    })();\n",
" </script>\n",
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  ]
}

```

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    },  
    "metadata": {},  
    "execution_count": 24  
  }  
],  
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" 3 Dependents   7043 non-null object\n",
" 4 tenure       7043 non-null int64\n",
" 5 PhoneService 7043 non-null object\n",
" 6 MultipleLines 7043 non-null object\n",
" 7 InternetService 7043 non-null object\n",
" 8 OnlineSecurity 7043 non-null object\n",
" 9 OnlineBackup 7043 non-null object\n",
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"12 StreamingTV   7043 non-null object\n",
"13 StreamingMovies 7043 non-null object\n",
"14 Contract      7043 non-null object\n",
"15 PaperlessBilling 7043 non-null object\n",
"16 PaymentMethod 7043 non-null object\n",
"17 MonthlyCharges 7043 non-null float64\n",
"18 TotalCharges  7043 non-null float64\n",
"19 Churn         7043 non-null object\n",
"dtypes: float64(2), int64(2), object(16)\n",
"memory usage: 1.1+ MB\n"
]
}
]
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"# checking the class distribution of target column\n",
"print(df[\"Churn\"].value_counts())"
],

```



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      "Yes  1869\n",
      "Name: count, dtype: int64\n"
    ]
  }
],
{
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    "1. Customer ID removed as it is not required for modelling\n",
    "2. No mmissing values in the dataset\n",
    "3. Missing values in the TotalCharges column were replaced with 0\n",
    "4. Class imbalance identified in the target"
  ],
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```

```
},
{
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    "**3. Exploratory Data Analysis (EDA)**"
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      "data": {
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    }
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```

```

    }
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    "outputId": "c2662004-a0b2-40cb-da55-512846470bc6"
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      "data": {
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          "      'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity',\n",
          "      'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV',\n",
          "      'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod',\n",
          "      'MonthlyCharges', 'TotalCharges', 'Churn'],\n",
          "      dtype='object')]"
        ]
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      "metadata": {}
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]

```

```

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          "  gender SeniorCitizen Partner Dependents tenure PhoneService \\n",
          "0 Female      0  Yes    No    1      No  \\n",
          "1 Male        0  No     No   34     Yes  \\n",
          "\\n",
          "  MultipleLines InternetService OnlineSecurity OnlineBackup \\n",
          "0 No phone service      DSL      No      Yes  \\n",
          "1      No      DSL      Yes      No  \\n",
          "\\n",
          "  DeviceProtection TechSupport StreamingTV StreamingMovies  Contr
act \\n",
          "0      No      No      No      No Month-to-month  \\n",
          "1      Yes      No      No      No  One year  \\n",
          "\\n",

```

```

" PaperlessBilling   PaymentMethod MonthlyCharges TotalCharges Chur
n \n",
"0      Yes Electronic check      29.85      29.85 No \n",
"1      No   Mailed check      56.95      1889.50 No "
],
"text/html": [
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-df-container\">\n",
  " <div>\n",
  "<style scoped>\n",
  " .dataframe tbody tr th:only-of-type {\n",
  "   vertical-align: middle;\n",
  " } \n",
  "\n",
  " .dataframe tbody tr th {\n",
  "   vertical-align: top;\n",
  " } \n",
  "\n",
  " .dataframe thead th {\n",
  "   text-align: right;\n",
  " } \n",
  "</style>\n",
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  " <thead>\n",
  " <tr style=\"text-align: right;\">\n",
  " <th></th>\n",
  " <th>gender</th>\n",
  " <th>SeniorCitizen</th>\n",
  " <th>Partner</th>\n",
  " <th>Dependents</th>\n",
  " <th>tenure</th>\n",
  " <th>PhoneService</th>\n",

```

```
" <th>MultipleLines</th>\n",
" <th>InternetService</th>\n",
" <th>OnlineSecurity</th>\n",
" <th>OnlineBackup</th>\n",
" <th>DeviceProtection</th>\n",
" <th>TechSupport</th>\n",
" <th>StreamingTV</th>\n",
" <th>StreamingMovies</th>\n",
" <th>Contract</th>\n",
" <th>PaperlessBilling</th>\n",
" <th>PaymentMethod</th>\n",
" <th>MonthlyCharges</th>\n",
" <th>TotalCharges</th>\n",
" <th>Churn</th>\n",
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" </thead>\n",
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" <td>0</td>\n",
" <td>Yes</td>\n",
" <td>No</td>\n",
" <td>1</td>\n",
" <td>No</td>\n",
" <td>No phone service</td>\n",
" <td>DSL</td>\n",
" <td>No</td>\n",
" <td>Yes</td>\n",
" <td>No</td>\n",
" <td>No</td>\n",
" <td>No</td>\n",
" <td>No</td>\n"
```

```
" <td>Month-to-month</td>\n",
" <td>Yes</td>\n",
" <td>Electronic check</td>\n",
" <td>29.85</td>\n",
" <td>29.85</td>\n",
" <td>No</td>\n",
" </tr>\n",
" <tr>\n",
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" <td>No</td>\n",
" <td>No</td>\n",
" <td>34</td>\n",
" <td>Yes</td>\n",
" <td>No</td>\n",
" <td>DSL</td>\n",
" <td>Yes</td>\n",
" <td>No</td>\n",
" <td>Yes</td>\n",
" <td>No</td>\n",
" <td>No</td>\n",
" <td>No</td>\n",
" <td>One year</td>\n",
" <td>No</td>\n",
" <td>Mailed check</td>\n",
" <td>56.95</td>\n",
" <td>1889.50</td>\n",
" <td>No</td>\n",
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"</table>\n",
"</div>\n",
```

```
" <div class=\"colab-df-buttons\">\n",
"\n",
" <div class=\"colab-df-container\">\n",
" <button class=\"colab-df-convert\" onclick=\"convertToInteractive('df-9
538c841-ea88-425a-9c02-248fb9db815a')\" \n",
"     title=\"Convert this dataframe to an interactive table.\" \n",
"     style=\"display:none;\">\n",
"\n",
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\"0 960 960 960\">\n",
" <path d=\"M120 120v720h720v720H120Zm60 500h600v160H180v1
60Zm220 220h160v160H400v160Zm0 220h160v160H400v160ZM180 400h1
60v160H180v160Zm440 0h160v160H620v160ZM180 180h160v160H180v16
0Zm440 0h160v160H620v160Z\"/>\n",
" </svg>\n",
" </button>\n",
"\n",
" <style>\n",
" .colab-df-container {\n",
"   display: flex;\n",
"   gap: 12px;\n",
" }\n",
"\n",
" .colab-df-convert {\n",
"   background-color: #E8F0FE;\n",
"   border: none;\n",
"   border-radius: 50%;\n",
"   cursor: pointer;\n",
"   display: none;\n",
"   fill: #1967D2;\n",
"   height: 32px;\n",
"   padding: 0 0 0 0;\n",
"   width: 32px;\n",
```



```
" }\n",
"\n",
" .colab-df-convert:hover {\n",
"   background-color: #E2EBFA;\n",
"   box-shadow: 0px 1px 2px rgba(60, 64, 67, 0.3), 0px 1px 3px 1px rgba(60, 64, 67, 0.15);\n",
"   fill: #174EA6;\n",
" }\n",
"\n",
" .colab-df-buttons div {\n",
"   margin-bottom: 4px;\n",
" }\n",
"\n",
" [theme=dark] .colab-df-convert {\n",
"   background-color: #3B4455;\n",
"   fill: #D2E3FC;\n",
" }\n",
"\n",
" [theme=dark] .colab-df-convert:hover {\n",
"   background-color: #434B5C;\n",
"   box-shadow: 0px 1px 3px 1px rgba(0, 0, 0, 0.15);\n",
"   filter: drop-shadow(0px 1px 2px rgba(0, 0, 0, 0.3));\n",
"   fill: #FFFFFF;\n",
" }\n",
" </style>\n",
"\n",
" <script>\n",
"   const buttonEl =\n",
"     document.querySelector('#df-9538c841-ea88-425a-9c02-248fb9d\nb815a button.colab-df-convert');\n",
"   buttonEl.style.display =\n",
"     google.colab.kernel.accessAllowed ? 'block' : 'none';\n",
"
```

```

"    async function convertToInteractive(key) {\n",
"        const element = document.querySelector('#df-9538c841-ea88-425a-9c02-248fb9db815a');\n",
"        const dataTable =\n",
"            await google.colab.kernel.invokeFunction('convertToInteractive',\n",
"                [key], {});\n",
"        if (!dataTable) return;\n",
"\n",
"        const docLinkHtml = 'Like what you see? Visit the ' +\n",
"            '<a target=\"_blank\" href=https://colab.research.google.com/notebooks/data_table.ipynb>data table notebook</a>'\n",
"            + ' to learn more about interactive tables.';\n",
"        element.innerHTML = \";\n",
"        dataTable['output_type'] = 'display_data';\n",
"        await google.colab.output.renderOutput(dataTable, element);\n",
"        const docLink = document.createElement('div');\n",
"        docLink.innerHTML = docLinkHtml;\n",
"        element.appendChild(docLink);\n",
"    }\n",
" </script>\n",
" </div>\n",
"\n",
"\n",
"<div id=\"df-58a50826-4bba-4f32-9090-143468305cae\">\n",
"  <button class=\"colab-df-quickchart\" onclick=\"quickchart('df-58a50826-4bba-4f32-9090-143468305cae')\"\n",
"    title=\"Suggest charts\"\n",
"    style=\"display:none;\">\n",
"\n",
"<svg xmlns=\"http://www.w3.org/2000/svg\" height=\"24px\" viewBox=\"0 0 24 24\"\n",
"  width=\"24px\">\n",
"  <g>\n",

```

```
"    <path d=\"M19 3H5c-1.1 0-2 .9-2 2v14c0 1.1.9 2 2 2h14c1.1 0 2-.9 2-2V
5c0-1.1-.9-2-2-2zM9 17H7v-7h2v7zm4 0h-2V7h2v10zm4 0h-2v-4h2v4z\"/>\n",
"  </g>\n",
"</svg>\n",
" </button>\n",
"\n",
"<style>\n",
"  .colab-df-quickchart {\n",
"    --bg-color: #E8F0FE;\n",
"    --fill-color: #1967D2;\n",
"    --hover-bg-color: #E2EBFA;\n",
"    --hover-fill-color: #174EA6;\n",
"    --disabled-fill-color: #AAA;\n",
"    --disabled-bg-color: #DDD;\n",
"  }\n",
"\n",
"  [theme=dark] .colab-df-quickchart {\n",
"    --bg-color: #3B4455;\n",
"    --fill-color: #D2E3FC;\n",
"    --hover-bg-color: #434B5C;\n",
"    --hover-fill-color: #FFFFFF;\n",
"    --disabled-bg-color: #3B4455;\n",
"    --disabled-fill-color: #666;\n",
"  }\n",
"\n",
"  .colab-df-quickchart {\n",
"    background-color: var(--bg-color);\n",
"    border: none;\n",
"    border-radius: 50%;\n",
"    cursor: pointer;\n",
"    display: none;\n",
"    fill: var(--fill-color);\n",
"    height: 32px;\n",

```

```
" padding: 0;\n",  
" width: 32px;\n",  
" }\n",  
"\n",  
" .colab-df-quickchart:hover {\n",  
" background-color: var(--hover-bg-color);\n",  
" box-shadow: 0 1px 2px rgba(60, 64, 67, 0.3), 0 1px 3px 1px rgba(60, 64,  
67, 0.15);\n",  
" fill: var(--button-hover-fill-color);\n",  
" }\n",  
"\n",  
" .colab-df-quickchart-complete:disabled,\n",  
" .colab-df-quickchart-complete:disabled:hover {\n",  
" background-color: var(--disabled-bg-color);\n",  
" fill: var(--disabled-fill-color);\n",  
" box-shadow: none;\n",  
" }\n",  
"\n",  
" .colab-df-spinner {\n",  
" border: 2px solid var(--fill-color);\n",  
" border-color: transparent;\n",  
" border-bottom-color: var(--fill-color);\n",  
" animation:\n",  
" spin 1s steps(1) infinite;\n",  
" }\n",  
"\n",  
" @keyframes spin {\n",  
" 0% {\n",  
" border-color: transparent;\n",  
" border-bottom-color: var(--fill-color);\n",  
" border-left-color: var(--fill-color);\n",  
" }\n",  
" 20% {\n",
```

```
" border-color: transparent;\n",
" border-left-color: var(--fill-color);\n",
" border-top-color: var(--fill-color);\n",
" }\n",
" 30% {\n",
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" border-top-color: var(--fill-color);\n",
" border-right-color: var(--fill-color);\n",
" }\n",
" 40% {\n",
" border-color: transparent;\n",
" border-right-color: var(--fill-color);\n",
" border-top-color: var(--fill-color);\n",
" }\n",
" 60% {\n",
" border-color: transparent;\n",
" border-right-color: var(--fill-color);\n",
" }\n",
" 80% {\n",
" border-color: transparent;\n",
" border-right-color: var(--fill-color);\n",
" border-bottom-color: var(--fill-color);\n",
" }\n",
" 90% {\n",
" border-color: transparent;\n",
" border-bottom-color: var(--fill-color);\n",
" }\n",
" }\n",
"</style>\n",
"\n",
" <script>\n",
" async function quickchart(key) {\n",
```

```

"    const quickchartButtonEl =\n",
"    document.querySelector('#' + key + ' button');\n",
"    quickchartButtonEl.disabled = true; // To prevent multiple clicks.\n",
"    quickchartButtonEl.classList.add('colab-df-spinner');\n",
"    try {\n",
"        const charts = await google.colab.kernel.invokeFunction(\n",
"            'suggestCharts', [key], {});\n",
"    } catch (error) {\n",
"        console.error('Error during call to suggestCharts:', error);\n",
"    }\n",
"    quickchartButtonEl.classList.remove('colab-df-spinner');\n",
"    quickchartButtonEl.classList.add('colab-df-quickchart-complete');\n",
"  }\n",
"  (() => {\n",
"    let quickchartButtonEl =\n",
"      document.querySelector('#df-58a50826-4bba-4f32-9090-143468305cae button');\n",
"    quickchartButtonEl.style.display =\n",
"      google.colab.kernel.accessAllowed ? 'block' : 'none';\n",
"  })();\n",
"</script>\n",
"</div>\n",
"\n",
"</div>\n",
"</div>\n",
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```

```

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```

```
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```

]
},
{
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      "height": 300
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          "count  7043.000000  7043.000000   7043.000000  7043.000000\n",
          "mean    0.162147   32.371149    64.761692  2279.734304\n",
          "std     0.368612   24.559481    30.090047  2266.794470\n",
          "min     0.000000    0.000000    18.250000    0.000000\n",
          "25%     0.000000    9.000000    35.500000   398.550000\n",
          "50%     0.000000   29.000000    70.350000  1394.550000\n",
          "75%     0.000000   55.000000    89.850000  3786.600000\n",
          "max     1.000000   72.000000   118.750000  8684.800000"
        ],
        "text/html": [
          "\n",

```

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ab-df-container\">\n",
" <div>\n",
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" .dataframe tbody tr th:only-of-type {\n",
"   vertical-align: middle;\n",
" } \n",
"\n",
" .dataframe tbody tr th {\n",
"   vertical-align: top;\n",
" } \n",
"\n",
" .dataframe thead th {\n",
"   text-align: right;\n",
" } \n",
"</style>\n",
"<table border=\"1\" class=\"dataframe\">\n",
" <thead>\n",
" <tr style=\"text-align: right;\">\n",
" <th></th>\n",
" <th>SeniorCitizen</th>\n",
" <th>tenure</th>\n",
" <th>MonthlyCharges</th>\n",
" <th>TotalCharges</th>\n",
" </tr>\n",
" </thead>\n",
" <tbody>\n",
" <tr>\n",
" <th>count</th>\n",
" <td>7043.000000</td>\n",
" <td>7043.000000</td>\n",
" <td>7043.000000</td>\n",
" <td>7043.000000</td>
```

```
" </tr>\n",
" <tr>\n",
"   <th>mean</th>\n",
"   <td>0.162147</td>\n",
"   <td>32.371149</td>\n",
"   <td>64.761692</td>\n",
"   <td>2279.734304</td>\n",
" </tr>\n",
" <tr>\n",
"   <th>std</th>\n",
"   <td>0.368612</td>\n",
"   <td>24.559481</td>\n",
"   <td>30.090047</td>\n",
"   <td>2266.794470</td>\n",
" </tr>\n",
" <tr>\n",
"   <th>min</th>\n",
"   <td>0.000000</td>\n",
"   <td>0.000000</td>\n",
"   <td>18.250000</td>\n",
"   <td>0.000000</td>\n",
" </tr>\n",
" <tr>\n",
"   <th>25%</th>\n",
"   <td>0.000000</td>\n",
"   <td>9.000000</td>\n",
"   <td>35.500000</td>\n",
"   <td>398.550000</td>\n",
" </tr>\n",
" <tr>\n",
"   <th>50%</th>\n",
"   <td>0.000000</td>\n",
"   <td>29.000000</td>
```

```

"    <td>70.350000</td>\n",
"    <td>1394.550000</td>\n",
"  </tr>\n",
"  <tr>\n",
"    <th>75%</th>\n",
"    <td>0.000000</td>\n",
"    <td>55.000000</td>\n",
"    <td>89.850000</td>\n",
"    <td>3786.600000</td>\n",
"  </tr>\n",
"  <tr>\n",
"    <th>max</th>\n",
"    <td>1.000000</td>\n",
"    <td>72.000000</td>\n",
"    <td>118.750000</td>\n",
"    <td>8684.800000</td>\n",
"  </tr>\n",
" </tbody>\n",
"</table>\n",
"</div>\n",
"  <div class=\"colab-df-buttons\">\n",
"\n",
"  <div class=\"colab-df-container\">\n",
"    <button class=\"colab-df-convert\" onclick=\"convertToInteractive('df-a
d64676b-842e-489e-bb54-6b3b8bc28bb0')\" \n",
"      title=\"Convert this dataframe to an interactive table.\" \n",
"      style=\"display:none;\">\n",
"\n",
"    <svg xmlns=\"http://www.w3.org/2000/svg\" height=\"24px\" viewBox=
\"0 960 960 960\">\n",
"      <path d=\"M120-120v-720h720v720H120Zm60-500h600v-160H180v1
60Zm220 220h160v-160H400v160Zm0 220h160v-160H400v160ZM180-400h1
60v-160H180v160Zm440 0h160v-160H620v160ZM180-180h160v-160H180v16

```

```
0Zm440 0h160v-160H620v160Z\"/>
" </svg>\n",
" </button>\n",
"\n",
" <style>\n",
" .colab-df-container {\n",
"   display: flex;\n",
"   gap: 12px;\n",
" } \n",
"\n",
" .colab-df-convert {\n",
"   background-color: #E8F0FE;\n",
"   border: none;\n",
"   border-radius: 50%;\n",
"   cursor: pointer;\n",
"   display: none;\n",
"   fill: #1967D2;\n",
"   height: 32px;\n",
"   padding: 0 0 0 0;\n",
"   width: 32px;\n",
" } \n",
"\n",
" .colab-df-convert:hover {\n",
"   background-color: #E2EBFA;\n",
"   box-shadow: 0px 1px 2px rgba(60, 64, 67, 0.3), 0px 1px 3px 1px rgba(60, 64, 67, 0.15);\n",
"   fill: #174EA6;\n",
" } \n",
"\n",
" .colab-df-buttons div {\n",
"   margin-bottom: 4px;\n",
" } \n",
"\n",
```

```

" [theme=dark] .colab-df-convert {\n",
"   background-color: #3B4455;\n",
"   fill: #D2E3FC;\n",
" } \n",
"\n",
" [theme=dark] .colab-df-convert:hover {\n",
"   background-color: #434B5C;\n",
"   box-shadow: 0px 1px 3px 1px rgba(0, 0, 0, 0.15);\n",
"   filter: drop-shadow(0px 1px 2px rgba(0, 0, 0, 0.3));\n",
"   fill: #FFFFFF;\n",
" } \n",
" </style>\n",
"\n",
" <script>\n",
"   const buttonEl =\n",
"     document.querySelector('#df-ad64676b-842e-489e-bb54-6b3b8b
c28bb0 button.colab-df-convert');\n",
"   buttonEl.style.display =\n",
"     google.colab.kernel.accessAllowed ? 'block' : 'none';\n",
"\n",
"   async function convertToInteractive(key) {\n",
"     const element = document.querySelector('#df-ad64676b-842e-489
e-bb54-6b3b8bc28bb0');\n",
"     const dataTable =\n",
"       await google.colab.kernel.invokeFunction('convertToInteractive',\n",
"         [key], {});\n",
"     if (!dataTable) return;\n",
"\n",
"     const docLinkHtml = 'Like what you see? Visit the ' +\n",
"       '<a target=\"_blank\" href=https://colab.research.google.com/noteb
ooks/data_table.ipynb>data table notebook</a>'\n",
"       + ' to learn more about interactive tables.';\n",
"     element.innerHTML = ";\n",

```

```

"    dataTable['output_type'] = 'display_data';\n",
"    await google.colab.output.renderOutput(dataTable, element);\n",
"    const docLink = document.createElement('div');\n",
"    docLink.innerHTML = docLinkHtml;\n",
"    element.appendChild(docLink);\n",
"  }\n",
"  </script>\n",
" </div>\n",
"\n",
"\n",
"<div id=\"df-7d1c4a7d-3213-4dee-944a-2cc1cd4755a0\">\n",
"  <button class=\"colab-df-quickchart\" onclick=\"quickchart('df-7d1c4a7d-3213-4dee-944a-2cc1cd4755a0')\" \n",
"    title=\"Suggest charts\" \n",
"    style=\"display:none;\">\n",
"\n",
"  <svg xmlns=\"http://www.w3.org/2000/svg\" height=\"24px\" viewBox=\"0 0 24 24\" \n",
"    width=\"24px\">\n",
"    <g>\n",
"      <path d=\"M19 3H5c-1.1 0-.9-.2 2v14c0 1.1.9 2 2 2h14c1.1 0 2-.9 2-2V5c0-1.1-.9-2-.9-2-2 2zM9 17H7v-7h2v7zm4 0h-2V7h2v10zm4 0h-2v-4h2v4z\"/>\n",
"    </g>\n",
"  </svg>\n",
"  </button>\n",
"\n",
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"    --bg-color: #E8F0FE;\n",
"    --fill-color: #1967D2;\n",
"    --hover-bg-color: #E2EBFA;\n",
"    --hover-fill-color: #174EA6;\n",
"    --disabled-fill-color: #AAA;\n",

```

```
"  --disabled-bg-color: #DDD;\n",
" }\n",
"\n",
" [theme=dark] .colab-df-quickchart {\n",
"  --bg-color: #3B4455;\n",
"  --fill-color: #D2E3FC;\n",
"  --hover-bg-color: #434B5C;\n",
"  --hover-fill-color: #FFFFFF;\n",
"  --disabled-bg-color: #3B4455;\n",
"  --disabled-fill-color: #666;\n",
" }\n",
"\n",
" .colab-df-quickchart {\n",
"  background-color: var(--bg-color);\n",
"  border: none;\n",
"  border-radius: 50%;\n",
"  cursor: pointer;\n",
"  display: none;\n",
"  fill: var(--fill-color);\n",
"  height: 32px;\n",
"  padding: 0;\n",
"  width: 32px;\n",
" }\n",
"\n",
" .colab-df-quickchart:hover {\n",
"  background-color: var(--hover-bg-color);\n",
"  box-shadow: 0 1px 2px rgba(60, 64, 67, 0.3), 0 1px 3px 1px rgba(60, 64, 67, 0.15);\n",
"  fill: var(--button-hover-fill-color);\n",
" }\n",
"\n",
" .colab-df-quickchart-complete:disabled,\n",
" .colab-df-quickchart-complete:disabled:hover {\n",

```



```
" background-color: var(--disabled-bg-color);\n",\n" fill: var(--disabled-fill-color);\n",\n" box-shadow: none;\n",\n" }\n",\n"\n",\n" .colab-df-spinner {\n",\n" border: 2px solid var(--fill-color);\n",\n" border-color: transparent;\n",\n" border-bottom-color: var(--fill-color);\n",\n" animation:\n",\n"   spin 1s steps(1) infinite;\n",\n" }\n",\n"\n",\n" @keyframes spin {\n",\n"   0% {\n",\n"     border-color: transparent;\n",\n"     border-bottom-color: var(--fill-color);\n",\n"     border-left-color: var(--fill-color);\n",\n"   }\n",\n"   20% {\n",\n"     border-color: transparent;\n",\n"     border-left-color: var(--fill-color);\n",\n"     border-top-color: var(--fill-color);\n",\n"   }\n",\n"   30% {\n",\n"     border-color: transparent;\n",\n"     border-left-color: var(--fill-color);\n",\n"     border-top-color: var(--fill-color);\n",\n"     border-right-color: var(--fill-color);\n",\n"   }\n",\n"   40% {\n",\n"     border-color: transparent;\n",\n"     border-right-color: var(--fill-color);\n",\n"
```

```
" border-top-color: var(--fill-color);\n",\n"}\n",\n" 60% {\n",\n" border-color: transparent;\n",\n" border-right-color: var(--fill-color);\n",\n"}\n",\n" 80% {\n",\n" border-color: transparent;\n",\n" border-right-color: var(--fill-color);\n",\n" border-bottom-color: var(--fill-color);\n",\n"}\n",\n" 90% {\n",\n" border-color: transparent;\n",\n" border-bottom-color: var(--fill-color);\n",\n"}\n",\n"}\n",\n"</style>\n",\n"\n",\n" <script>\n",\n" async function quickchart(key) {\n",\n" const quickchartButtonEl =\n",\n" document.querySelector('#' + key + ' button');\n",\n" quickchartButtonEl.disabled = true; // To prevent multiple clicks.\n",\n" quickchartButtonEl.classList.add('colab-df-spinner');\n",\n" try {\n",\n" const charts = await google.colab.kernel.invokeFunction(\n",\n" 'suggestCharts', [key], {});\n",\n" } catch (error) {\n",\n" console.error('Error during call to suggestCharts:', error);\n",\n" }\n",\n" quickchartButtonEl.classList.remove('colab-df-spinner');\n",\n" quickchartButtonEl.classList.add('colab-df-quickchart-complete');\n",\n" }\n",
```

```

"  (() => {\n",
"    let quickchartButtonEl =\n",
"      document.querySelector('#df-7d1c4a7d-3213-4dee-944a-2cc1cd47
55a0 button');\n",
"    quickchartButtonEl.style.display =\n",
"      google.colab.kernel.accessAllowed ? 'block' : 'none';\n",
"  })();\n",
" </script>\n",
"</div>\n",
"\n",
" </div>\n",
" </div>\n"
],
"application/vnd.google.colaboratory.intrinsic+json": {
  "type": "dataframe",
  "summary": "{\n  \"name\": \"df\",\n  \"rows\": 8,\n  \"fields\": [\n    {\n      \"column\": \"SeniorCitizen\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 2489.9992387084,\n        \"min\": 0.0,\n        \"max\": 7043.0,\n        \"num_unique_values\": 5,\n        \"samples\": [\n          0.1621468124378816,\n          1.0,\n          0.36861160561002687\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"tenure\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 2478.9752758409018,\n        \"min\": 0.0,\n        \"max\": 7043.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          32.37114865824223,\n          29.0,\n          7043.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"MonthlyCharges\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 2468.7047672837775,\n        \"min\": 18.25,\n        \"max\": 7043.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          64.76169246059918,\n          70.35,\n          7043.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"TotalCharges\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 3122.5732655623974,\n        \"min\": 0.0,\n        \"max\": 8684.8,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          2279.7343035638223,\n          1394.55,\n          7043.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    }\n  ]\n}"

```

```

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  },
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]
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  }
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    "\n",
    "    plt.figure(figsize=(5, 3))\n",
    "    sns.histplot(df[column_name], kde=True)\n",
    "    plt.title(f\"Distribution of {column_name}\")\n",

```

```

"\n",
" # calculate the mean and median values for the columns\n",
" col_mean = df[column_name].mean()\n",
" col_median = df[column_name].median()\n",
"\n",
" # add vertical lines for mean and median\n",
" plt.axvline(col_mean, color=\"red\", linestyle=\"--\", label=\"Mean\")\n",
" plt.axvline(col_median, color=\"green\", linestyle=\"-\", label=\"Median\")\n",
"\n",
" plt.legend()\n",
"\n",
" plt.show()"
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{
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    "plot_histogram(df, \"tenure\")"
  ],
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    "id": "C-ii594gRLiv",
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  },
  "execution_count": 36,

```

```
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```

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Box plot for numerical features

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==\n"

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```

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      "0 Female      0  Yes    No    1      No  \\n",
      "1 Male        0  No     No   34     Yes  \\n",
      "2 Male        0  No     No    2     Yes  \\n",
      "\\n",
      "  MultipleLines InternetService OnlineSecurity OnlineBackup \\n",
      "0 No phone service      DSL      No      Yes  \\n",
      "1      No      DSL      Yes      No  \\n",
      "2      No      DSL      Yes      Yes  \\n",
      "\\n",
      "  DeviceProtection TechSupport StreamingTV StreamingMovies  Contr
act \\n",
      "0      No      No      No      No  Month-to-month  \\n",
      "1      Yes     No      No      No   One year  \\n",
      "2      No      No      No      No  Month-to-month  \\n",
      "\\n",
      "  PaperlessBilling  PaymentMethod MonthlyCharges TotalCharges Chur
n  \\n",
      "0      Yes Electronic check      29.85      29.85  No  \\n",
      "1      No   Mailed check      56.95      1889.50  No  \\n",
      "2      Yes   Mailed check      53.85      108.15  Yes  "
    ],
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-df-container\">\\n",

```

```
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" .dataframe tbody tr th:only-of-type {\n",
"     vertical-align: middle;\n",
" } \n",
"\n",
" .dataframe tbody tr th {\n",
"     vertical-align: top;\n",
" } \n",
"\n",
" .dataframe thead th {\n",
"     text-align: right;\n",
" } \n",
"</style>\n",
"<table border=\"1\" class=\"dataframe\">\n",
" <thead>\n",
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"     <th>gender</th>\n",
"     <th>SeniorCitizen</th>\n",
"     <th>Partner</th>\n",
"     <th>Dependents</th>\n",
"     <th>tenure</th>\n",
"     <th>PhoneService</th>\n",
"     <th>MultipleLines</th>\n",
"     <th>InternetService</th>\n",
"     <th>OnlineSecurity</th>\n",
"     <th>OnlineBackup</th>\n",
"     <th>DeviceProtection</th>\n",
"     <th>TechSupport</th>\n",
"     <th>StreamingTV</th>\n",
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"     <th>Contract</th>
```

```
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" <td>No</td>\n",
" <td>No</td>\n",
" <td>No</td>\n",
" <td>No</td>\n",
" <td>Month-to-month</td>\n",
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" <td>29.85</td>\n",
" <td>No</td>\n",
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" <tr>\n",
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" <td>No</td>\n",
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" <td>One year</td>\n",
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" <td>Mailed check</td>\n",
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" <td>1889.50</td>\n",
" <td>No</td>\n",
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" <td>No</td>\n",
" <td>DSL</td>\n",
" <td>Yes</td>\n",
" <td>Yes</td>\n",

```

"    <td>No</td>\n",
"    <td>No</td>\n",
"    <td>No</td>\n",
"    <td>No</td>\n",
"    <td>Month-to-month</td>\n",
"    <td>Yes</td>\n",
"    <td>Mailed check</td>\n",
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"    <td>108.15</td>\n",
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"  <div class=\"colab-df-container\">\n",
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e03b093-4792-4225-8591-b7be119d018e')\">\n",
"      title=\"Convert this dataframe to an interactive table.\"\n",
"      style=\"display:none;\"\n",
"\n",
"    <svg xmlns=\"http://www.w3.org/2000/svg\" height=\"24px\" viewBox=
\"0 -960 960 960\">\n",
"      <path d=\"M120-120v-720h720v720H120Zm60-500h600v-160H180v1
60Zm220 220h160v-160H400v160Zm0 220h160v-160H400v160ZM180-400h1
60v-160H180v160Zm440 0h160v-160H620v160ZM180-180h160v-160H180v16
0Zm440 0h160v-160H620v160Z\"/>\n",
"    </svg>\n",
"  </button>\n",
"\n",
"  <style>\n",
"    .colab-df-container {\n",

```



```
"  display:flex;\n",
"  gap: 12px;\n",
" }\n",
"\n",
" .colab-df-convert {\n",
"  background-color: #E8F0FE;\n",
"  border: none;\n",
"  border-radius: 50%;\n",
"  cursor: pointer;\n",
"  display: none;\n",
"  fill: #1967D2;\n",
"  height: 32px;\n",
"  padding: 0 0 0 0;\n",
"  width: 32px;\n",
" }\n",
"\n",
" .colab-df-convert:hover {\n",
"  background-color: #E2EBFA;\n",
"  box-shadow: 0px 1px 2px rgba(60, 64, 67, 0.3), 0px 1px 3px 1px rgba(60, 64, 67, 0.15);\n",
"  fill: #174EA6;\n",
" }\n",
"\n",
" .colab-df-buttons div {\n",
"  margin-bottom: 4px;\n",
" }\n",
"\n",
" [theme=dark] .colab-df-convert {\n",
"  background-color: #3B4455;\n",
"  fill: #D2E3FC;\n",
" }\n",
"\n",
" [theme=dark] .colab-df-convert:hover {\n",
```

```

" background-color: #434B5C;\n",
" box-shadow: 0px 1px 3px 1px rgba(0, 0, 0, 0.15);\n",
" filter: drop-shadow(0px 1px 2px rgba(0, 0, 0, 0.3));\n",
" fill: #FFFFFF;\n",
" }\n",
" </style>\n",
"\n",
" <script>\n",
" const buttonEl =\n",
" document.querySelector('#df-be03b093-4792-4225-8591-b7be119d018e button.colab-df-convert');\n",
" buttonEl.style.display =\n",
" google.colab.kernel.accessAllowed ? 'block' : 'none';\n",
"\n",
" async function convertToInteractive(key) {\n",
" const element = document.querySelector('#df-be03b093-4792-4225-8591-b7be119d018e');\n",
" const dataTable =\n",
" await google.colab.kernel.invokeFunction('convertToInteractive',\n",
" [key], {});\n",
" if (!dataTable) return;\n",
"\n",
" const docLinkHtml = 'Like what you see? Visit the ' +\n",
" '<a target=\"_blank\" href=https://colab.research.google.com/notebooks/data_table.ipynb>data table notebook</a>'\n",
" + ' to learn more about interactive tables.';\n",
" element.innerHTML = \";\n",
" dataTable['output_type'] = 'display_data';\n",
" await google.colab.output.renderOutput(dataTable, element);\n",
" const docLink = document.createElement('div');\n",
" docLink.innerHTML = docLinkHtml;\n",
" element.appendChild(docLink);\n",
" }\n",

```

```
" </script>\n",
" </div>\n",
"\n",
"\n",
"<div id=\"df-42debdd9-64be-45fd-a64a-d6fc17aa711c\">\n",
" <button class=\"colab-df-quickchart\" onclick=\"quickchart('df-42debdd
9-64be-45fd-a64a-d6fc17aa711c')\">\n",
"     title=\"Suggest charts\"\n",
"     style=\"display:none;\">\n",
"\n",
"<svg xmlns=\"http://www.w3.org/2000/svg\" height=\"24px\"viewBox=\"
0 0 24 24\">\n",
"   width=\"24px\">\n",
"   <g>\n",
"     <path d=\"M19 3H5c-1.1 0-2 .9-2 2v14c0 1.1.9 2 2 2h14c1.1 0 2 .9 2 2V
5c0-1.1-.9-2-2-2zM9 17H7v-7h2v7zm4 0h-2V7h2v10zm4 0h-2v-4h2v4z\"/>\n",
"   </g>\n",
"</svg>\n",
"</button>\n",
"\n",
"<style>\n",
" .colab-df-quickchart {\n",
"   --bg-color: #E8F0FE;\n",
"   --fill-color: #1967D2;\n",
"   --hover-bg-color: #E2EBFA;\n",
"   --hover-fill-color: #174EA6;\n",
"   --disabled-fill-color: #AAA;\n",
"   --disabled-bg-color: #DDD;\n",
" }\n",
"\n",
" [theme=dark] .colab-df-quickchart {\n",
"   --bg-color: #3B4455;\n",
"   --fill-color: #D2E3FC;\n",

```

```
"  --hover-bg-color: #434B5C;\n",  
"  --hover-fill-color: #FFFFFF;\n",  
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"  --disabled-fill-color: #666;\n",  
" }\n",  
"\n",  
" .colab-df-quickchart {\n",  
"   background-color: var(--bg-color);\n",  
"   border: none;\n",  
"   border-radius: 50%;\n",  
"   cursor: pointer;\n",  
"   display: none;\n",  
"   fill: var(--fill-color);\n",  
"   height: 32px;\n",  
"   padding: 0;\n",  
"   width: 32px;\n",  
" }\n",  
"\n",  
" .colab-df-quickchart:hover {\n",  
"   background-color: var(--hover-bg-color);\n",  
"   box-shadow: 0 1px 2px rgba(60, 64, 67, 0.3), 0 1px 3px 1px rgba(60, 64,  
67, 0.15);\n",  
"   fill: var(--button-hover-fill-color);\n",  
" }\n",  
"\n",  
" .colab-df-quickchart-complete:disabled,\n",  
" .colab-df-quickchart-complete:disabled:hover {\n",  
"   background-color: var(--disabled-bg-color);\n",  
"   fill: var(--disabled-fill-color);\n",  
"   box-shadow: none;\n",  
" }\n",  
"\n",  
" .colab-df-spinner {\n",
```

```
" border: 2px solid var(--fill-color);\n",  
" border-color: transparent;\n",  
" border-bottom-color: var(--fill-color);\n",  
" animation:\n",  
"   spin 1s steps(1) infinite;\n",  
" }\n",  
"\n",  
" @keyframes spin {\n",  
"   0% {\n",  
"     border-color: transparent;\n",  
"     border-bottom-color: var(--fill-color);\n",  
"     border-left-color: var(--fill-color);\n",  
"   }\n",  
"   20% {\n",  
"     border-color: transparent;\n",  
"     border-left-color: var(--fill-color);\n",  
"     border-top-color: var(--fill-color);\n",  
"   }\n",  
"   30% {\n",  
"     border-color: transparent;\n",  
"     border-left-color: var(--fill-color);\n",  
"     border-top-color: var(--fill-color);\n",  
"     border-right-color: var(--fill-color);\n",  
"   }\n",  
"   40% {\n",  
"     border-color: transparent;\n",  
"     border-right-color: var(--fill-color);\n",  
"     border-top-color: var(--fill-color);\n",  
"   }\n",  
"   60% {\n",  
"     border-color: transparent;\n",  
"     border-right-color: var(--fill-color);\n",  
"   }\n",  
" }
```

```

" 80% {\n",
"   border-color: transparent;\n",
"   border-right-color: var(--fill-color);\n",
"   border-bottom-color: var(--fill-color);\n",
" }\n",
" 90% {\n",
"   border-color: transparent;\n",
"   border-bottom-color: var(--fill-color);\n",
" }\n",
" }\n",
"</style>\n",
"\n",
" <script>\n",
"   async function quickchart(key) {\n",
"     const quickchartButtonEl =\n",
"       document.querySelector('#' + key + ' button');\n",
"     quickchartButtonEl.disabled = true; // To prevent multiple clicks.\n",
"     quickchartButtonEl.classList.add('colab-df-spinner');\n",
"     try {\n",
"       const charts = await google.colab.kernel.invokeFunction(\n",
"         'suggestCharts', [key], {});\n",
"     } catch (error) {\n",
"       console.error('Error during call to suggestCharts:', error);\n",
"     }\n",
"     quickchartButtonEl.classList.remove('colab-df-spinner');\n",
"     quickchartButtonEl.classList.add('colab-df-quickchart-complete');\n",
"   }\n",
"   (() => {\n",
"     let quickchartButtonEl =\n",
"       document.querySelector('#df-42debdd9-64be-45fd-a64a-d6fc17aa\n711c button');\n",
"     quickchartButtonEl.style.display =\n",
"       google.colab.kernel.accessAllowed ? 'block' : 'none';

```

```

" }());\n",
" </script>\n",
"</div>\n",
"\n",
" </div>\n",
" </div>\n"
],
"application/vnd.google.colaboratory.intrinsic+json": {
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    \"num_unique_values\": 3,\n    \"samples\": [\n      \"No phone service\",\n      \"No\",\n      \"Yes\"\n    ],\n    \"semantic_type\": \"\",\n    \"description\": \"\"

```

```
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```



```

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    }
  }
}

```

```

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  ]
}
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```

```

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      "1  Male        0  No     No   34     Yes  \\n",
      "2  Male        0  No     No    2     Yes  \\n",
      "\\n",
      "  MultipleLines InternetService OnlineSecurity OnlineBackup \\n",
      "0 No phone service      DSL      No      Yes  \\n",
      "1      No      DSL      Yes      No  \\n",
      "2      No      DSL      Yes      Yes  \\n",
      "\\n",
      "  DeviceProtection TechSupport StreamingTV StreamingMovies  Contr
act \\n",
      "0      No      No      No      No Month-to-month  \\n",
      "1      Yes      No      No      No  One year  \\n",
      "2      No      No      No      No Month-to-month  \\n",
      "\\n",
      "  PaperlessBilling  PaymentMethod MonthlyCharges TotalCharges Chur
n  \\n",
      "0      Yes Electronic check      29.85      29.85      0  \\n",
      "1      No  Mailed check      56.95      1889.50      0  \\n",
      "2      Yes  Mailed check      53.85      108.15      1  "
    ],
    "text/html": [
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      "  <div id=\"df-9e73c193-e402-408b-9510-df41f1bc8305\" class=\"colab-
df-container\">\\n",
      "    <div>\\n",
      "<style scoped>\\n",

```

```
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"     vertical-align: middle;\n",
" } \n",
"\n",
" .dataframe tbody tr th {\n",
"     vertical-align: top;\n",
" } \n",
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" .dataframe thead th {\n",
"     text-align: right;\n",
" } \n",
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" <th>SeniorCitizen</th>\n",
" <th>Partner</th>\n",
" <th>Dependents</th>\n",
" <th>tenure</th>\n",
" <th>PhoneService</th>\n",
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" <th>OnlineSecurity</th>\n",
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" <th>DeviceProtection</th>\n",
" <th>TechSupport</th>\n",
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" <th>Contract</th>\n",
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" <td>No</td>\n",
" <td>No</td>\n",
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" <tr>\n",
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```

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" <td>One year</td>\n",
" <td>No</td>\n",
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" <td>Yes</td>\n",
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" <td>No</td>\n",

```

"    <td>No</td>\n",
"    <td>No</td>\n",
"    <td>Month-to-month</td>\n",
"    <td>Yes</td>\n",
"    <td>Mailed check</td>\n",
"    <td>53.85</td>\n",
"    <td>108.15</td>\n",
"    <td>1</td>\n",
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" </tbody>\n",
"</table>\n",
"</div>\n",
"  <div class=\"colab-df-buttons\">\n",
"\n",
"  <div class=\"colab-df-container\">\n",
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e73c193-e402-408b-9510-df41f1bc8305')\" \n",
"      title=\"Convert this dataframe to an interactive table.\" \n",
"      style=\"display:none;\">\n",
"\n",
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"      <path d=\"M120-120v-720h720v720H120Zm60-500h600v-160H180v1
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60v-160H180v160Zm440 0h160v-160H620v160ZM180-180h160v-160H180v16
0Zm440 0h160v-160H620v160Z\"/>\n",
"    </svg>\n",
"  </button>\n",
"\n",
"  <style>\n",
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"      display:flex;\n",
"      gap: 12px;\n",

```

```
" }\n",
"\n",
" .colab-df-convert {\n",
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"   border: none;\n",
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"   fill: #1967D2;\n",
"   height: 32px;\n",
"   padding: 0 0 0 0;\n",
"   width: 32px;\n",
" }\n",
"\n",
" .colab-df-convert:hover {\n",
"   background-color: #E2EBFA;\n",
"   box-shadow: 0px 1px 2px rgba(60, 64, 67, 0.3), 0px 1px 3px 1px rgba(60, 64, 67, 0.15);\n",
"   fill: #174EA6;\n",
" }\n",
"\n",
" .colab-df-buttons div {\n",
"   margin-bottom: 4px;\n",
" }\n",
"\n",
" [theme=dark] .colab-df-convert {\n",
"   background-color: #3B4455;\n",
"   fill: #D2E3FC;\n",
" }\n",
"\n",
" [theme=dark] .colab-df-convert:hover {\n",
"   background-color: #434B5C;\n",
"   box-shadow: 0px 1px 3px 1px rgba(0, 0, 0, 0.15);\n",
```



```

" filter: drop-shadow(0px 1px 2px rgba(0, 0, 0, 0.3));\n",
" fill: #FFFFFF;\n",
" }\n",
" </style>\n",
"\n",
" <script>\n",
"   const buttonEl =\n",
"     document.querySelector('#df-9e73c193-e402-408b-9510-df41f1bc8305 button.colab-df-convert');\n",
"   buttonEl.style.display =\n",
"     google.colab.kernel.accessAllowed ? 'block' : 'none';\n",
"\n",
"   async function convertToInteractive(key) {\n",
"     const element = document.querySelector('#df-9e73c193-e402-408b-9510-df41f1bc8305');\n",
"     const dataTable =\n",
"       await google.colab.kernel.invokeFunction('convertToInteractive',\n",
"         [key], {});\n",
"     if (!dataTable) return;\n",
"\n",
"     const docLinkHtml = 'Like what you see? Visit the ' +\n",
"       '<a target=\"_blank\" href=https://colab.research.google.com/notebooks/data_table.ipynb>data table notebook</a>'\n",
"       + ' to learn more about interactive tables.';\n",
"     element.innerHTML = \"\n",
"       dataTable['output_type'] = 'display_data';\n",
"       await google.colab.output.renderOutput(dataTable, element);\n",
"       const docLink = document.createElement('div');\n",
"       docLink.innerHTML = docLinkHtml;\n",
"       element.appendChild(docLink);\n",
"     }\n",
"   </script>\n",
" </div>\n",

```

```
"\n",
"\n",
"<div id=\"df-3d4be714-c13c-481b-9ade-7621f4856bdc\">\n",
"  <button class=\"colab-df-quickchart\" onclick=\"quickchart('df-3d4be71
4-c13c-481b-9ade-7621f4856bdc')\" \n",
"    title=\"Suggest charts\" \n",
"    style=\"display:none;\">\n",
"\n",
"<svg xmlns=\"http://www.w3.org/2000/svg\" height=\"24px\" viewBox=\"
0 0 24 24\" \n",
"  width=\"24px\">\n",
"  <g>\n",
"    <path d=\"M19 3H5c-1.1 0-2 .9-2 2v14c0 1.1 .9 2 2 2h14c1.1 0 2 .9 2 2V
5c0-1.1-.9-2-2-2zM9 17H7v-7h2v7zm4 0h-2V7h2v10zm4 0h-2v-4h2v4z\"/>\n",
"  </g>\n",
"</svg>\n",
"</button>\n",
"\n",
"<style>\n",
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"    --bg-color: #E8F0FE;\n",
"    --fill-color: #1967D2;\n",
"    --hover-bg-color: #E2EBFA;\n",
"    --hover-fill-color: #174EA6;\n",
"    --disabled-fill-color: #AAA;\n",
"    --disabled-bg-color: #DDD;\n",
"  }\n",
"\n",
"  [theme=dark] .colab-df-quickchart {\n",
"    --bg-color: #3B4455;\n",
"    --fill-color: #D2E3FC;\n",
"    --hover-bg-color: #434B5C;\n",
"    --hover-fill-color: #FFFFFF;\n",

```

```
"  --disabled-bg-color: #3B4455;\n",\n"  --disabled-fill-color: #666;\n",\n" }\n",\n"\n",\n" .colab-df-quickchart {\n",\n"  background-color: var(--bg-color);\n",\n"  border: none;\n",\n"  border-radius: 50%;\n",\n"  cursor: pointer;\n",\n"  display: none;\n",\n"  fill: var(--fill-color);\n",\n"  height: 32px;\n",\n"  padding: 0;\n",\n"  width: 32px;\n",\n" }\n",\n"\n",\n" .colab-df-quickchart:hover {\n",\n"  background-color: var(--hover-bg-color);\n",\n"  box-shadow: 0 1px 2px rgba(60, 64, 67, 0.3), 0 1px 3px 1px rgba(60, 64,\n67, 0.15);\n",\n"  fill: var(--button-hover-fill-color);\n",\n" }\n",\n"\n",\n" .colab-df-quickchart-complete:disabled,\n",\n" .colab-df-quickchart-complete:disabled:hover {\n",\n"  background-color: var(--disabled-bg-color);\n",\n"  fill: var(--disabled-fill-color);\n",\n"  box-shadow: none;\n",\n" }\n",\n"\n",\n" .colab-df-spinner {\n",\n"  border: 2px solid var(--fill-color);\n",\n"  border-color: transparent;\n",
```

```
" border-bottom-color: var(--fill-color);\n",\n" animation:\n",\n"   spin 1s steps(1) infinite;\n",\n" }\n",\n"\n",\n" @keyframes spin {\n",\n"   0% {\n",\n"     border-color: transparent;\n",\n"     border-bottom-color: var(--fill-color);\n",\n"     border-left-color: var(--fill-color);\n",\n"   }\n",\n"   20% {\n",\n"     border-color: transparent;\n",\n"     border-left-color: var(--fill-color);\n",\n"     border-top-color: var(--fill-color);\n",\n"   }\n",\n"   30% {\n",\n"     border-color: transparent;\n",\n"     border-left-color: var(--fill-color);\n",\n"     border-top-color: var(--fill-color);\n",\n"     border-right-color: var(--fill-color);\n",\n"   }\n",\n"   40% {\n",\n"     border-color: transparent;\n",\n"     border-right-color: var(--fill-color);\n",\n"     border-top-color: var(--fill-color);\n",\n"   }\n",\n"   60% {\n",\n"     border-color: transparent;\n",\n"     border-right-color: var(--fill-color);\n",\n"   }\n",\n"   80% {\n",\n"     border-color: transparent;
```

```

" border-right-color: var(--fill-color);\n",
" border-bottom-color: var(--fill-color);\n",
" }\n",
" 90% {\n",
" border-color: transparent;\n",
" border-bottom-color: var(--fill-color);\n",
" }\n",
" }\n",
"</style>\n",
"\n",
" <script>\n",
" async function quickchart(key) {\n",
" const quickchartButtonEl =\n",
" document.querySelector('#' + key + ' button');\n",
" quickchartButtonEl.disabled = true; // To prevent multiple clicks.\n",
" quickchartButtonEl.classList.add('colab-df-spinner');\n",
" try {\n",
" const charts = await google.colab.kernel.invokeFunction(\n",
" 'suggestCharts', [key], {});\n",
" } catch (error) {\n",
" console.error('Error during call to suggestCharts:', error);\n",
" }\n",
" quickchartButtonEl.classList.remove('colab-df-spinner');\n",
" quickchartButtonEl.classList.add('colab-df-quickchart-complete');\n",
" }\n",
" (() => {\n",
" let quickchartButtonEl =\n",
" document.querySelector('#df-3d4be714-c13c-481b-9ade-7621f485\n6bdc button');\n",
" quickchartButtonEl.style.display =\n",
" google.colab.kernel.accessAllowed ? 'block' : 'none';\n",
" })();\n",
" </script>\n",

```

```

"</div>\n",
"\n",
" </div>\n",
" </div>\n"
],
"application/vnd.google.colaboratory.intrinsic+json": {
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\n        \"num_unique_values\": 3,\n        \"samples\": [\n          \"No phone service\",\n          \"No\",\n          \"Yes\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"InternetService\",\n      \"properties\": {\n        \"dtype\": \"category\",
\n        \"num_unique_values\": 3,\n        \"samples\": [\n          \"DSL\",\n          \"Fibe

```

```
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```

```

    "properties": {\n      "dtype": "number",\n      "std": 2266.7944696890195,\n      "min": 0.0,\n      "max": 8684.8,\n      "num_unique_values": 6531,\n      "samples": [\n        4600.7,\n        20.35\n      ],\n      "semantic_type": "",\n      "description": ""\n    }\n  },\n  {\n    "column": "Churn",\n    "properties": {\n      "dtype": "number",\n      "std": 0,\n      "min": 0,\n      "max": 1,\n      "num_unique_values": 2,\n      "samples": [\n        1,\n        0\n      ],\n      "semantic_type": "",\n      "description": ""\n    }\n  }\n]\n}\n}\n},\n  "metadata": {},\n  "execution_count": 58\n}\n],\n},\n{\n  "cell_type": "code",\n  "source": [\n    "print(df[\"Churn\"].value_counts())"\n  ],\n  "metadata": {\n    "colab": {\n      "base_uri": "https://localhost:8080/"\n    },\n    "id": "mcgXdTCwYsnw",\n    "outputId": "e4c5bf05-b86d-4d4f-ddb7-983d53943f78"\n  },\n  "execution_count": 59,\n  "outputs": [\n    {\n      "output_type": "stream",\n      "name": "stdout",\n      "text": [\n
```



```

    "Churn\n",
    "0  5174\n",
    "1  1869\n",
    "Name: count, dtype: int64\n"
  ]
}
]
},
{
  "cell_type": "markdown",
  "source": [
    "Label encoding of categorical fetaures"
  ],
  "metadata": {
    "id": "Cs3j58UWY8zJ"
  }
},
{
  "cell_type": "code",
  "source": [
    "# identifying columns with object data type\n",
    "object_columns = df.select_dtypes(include=\"object\").columns"
  ],
  "metadata": {
    "id": "Q3-YraKcY4oC"
  },
  "execution_count": 60,
  "outputs": []
},
{
  "cell_type": "code",
  "source": [
    "print(object_columns)"
  ]
}

```

```

],
"metadata": {
  "colab": {
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  },
  "id": "EiFRA573ZNEy",
  "outputId": "c23a12d6-2c54-42a5-a6d0-bd84cc613716"
},
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"outputs": [
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    "name": "stdout",
    "text": [
      "Index(['gender', 'Partner', 'Dependents', 'PhoneService', 'MultipleLines',\n",
      "      'InternetService', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',\n",
      "      'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract',\n",
      "      'PaperlessBilling', 'PaymentMethod'],\n",
      "      dtype='object')\n"
    ]
  }
],
{
  "cell_type": "code",
  "source": [
    "# initialize a dictionary to save the encoders\n",
    "encoders = {}\n",
    "\n",
    "# apply label encoding and store the encoders\n",
    "for column in object_columns:\n",
    "    label_encoder = LabelEncoder()\n",
    "    df[column] = label_encoder.fit_transform(df[column])\n",

```

```

" encoders[column] = label_encoder\n",
"\n",
"\n",
"# save the encoders to a pickle file\n",
"with open(\"encoders.pkl\", \"wb\") as f:\n",
" pickle.dump(encoders, f)\n"
],
"metadata": {
  "id": "Sq6c2atrZOdu"
},
"execution_count": 62,
"outputs": []
},
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  "cell_type": "code",
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    "encoders"
  ],
  "metadata": {
    "colab": {
      "base_uri": "https://localhost:8080/"
    },
    "id": "RV5ff4YVZpEQ",
    "outputId": "1615b953-f57c-4f40-fe61-407e009d1c32"
  },
  "execution_count": 63,
  "outputs": [
    {
      "output_type": "execute_result",
      "data": {
        "text/plain": [
          "{ 'gender': LabelEncoder(),\n",
          " 'Partner': LabelEncoder(),\n",

```

```
"'Dependents': LabelEncoder(),\n",  
"'PhoneService': LabelEncoder(),\n",  
"'MultipleLines': LabelEncoder(),\n",  
"'InternetService': LabelEncoder(),\n",  
"'OnlineSecurity': LabelEncoder(),\n",  
"'OnlineBackup': LabelEncoder(),\n",  
"'DeviceProtection': LabelEncoder(),\n",  
"'TechSupport': LabelEncoder(),\n",  
"'StreamingTV': LabelEncoder(),\n",  
"'StreamingMovies': LabelEncoder(),\n",  
"'Contract': LabelEncoder(),\n",  
"'PaperlessBilling': LabelEncoder(),\n",  
"'PaymentMethod': LabelEncoder()}"  
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,  
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}  
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,  
{  
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  "source": [  
    "df.head()"  
  ],  
  "metadata": {  
    "colab": {  
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    },  
    "id": "km5wYiXAZsb0",  
    "outputId": "9ee8bb31-fabe-42b5-a927-92b8f914c54d"  
  },  
}
```

```

"execution_count": 64,
"outputs": [
{
  "output_type": "execute_result",
  "data": {
    "text/plain": [
      " gender SeniorCitizen Partner Dependents tenure PhoneService \\n",
      "0  0      0  1      0  1      0 \\n",
      "1  1      0  0      0 34      1 \\n",
      "2  1      0  0      0  2      1 \\n",
      "3  1      0  0      0 45      0 \\n",
      "4  0      0  0      0  2      1 \\n",
      "\\n",
      " MultipleLines InternetService OnlineSecurity OnlineBackup \\n",
      "0      1      0      0      2 \\n",
      "1      0      0      2      0 \\n",
      "2      0      0      2      2 \\n",
      "3      1      0      2      0 \\n",
      "4      0      1      0      0 \\n",
      "\\n",
      " DeviceProtection TechSupport StreamingTV StreamingMovies Contra
ct \\n",
      "0      0      0      0      0  0 \\n",
      "1      2      0      0      0  1 \\n",
      "2      0      0      0      0  0 \\n",
      "3      2      2      0      0  1 \\n",
      "4      0      0      0      0  0 \\n",
      "\\n",
      " PaperlessBilling PaymentMethod MonthlyCharges TotalCharges Churn
\\n",
      "0      1      2      29.85      29.85  0 \\n",
      "1      0      3      56.95     1889.50  0 \\n",
      "2      1      3      53.85     108.15  1 \\n",

```

```

"3      0      0      42.30      1840.75      0 \n",
"4      1      2      70.70      151.65      1 "
],
"text/html": [
  "\n",
  " <div id=\"df-709e34a1-db8d-4e06-81f9-17fe7bfb47cf\" class=\"colab-d
f-container\">\n",
  " <div>\n",
  "<style scoped>\n",
  " .dataframe tbody tr th:only-of-type {\n",
  "   vertical-align: middle;\n",
  " } \n",
  "\n",
  " .dataframe tbody tr th {\n",
  "   vertical-align: top;\n",
  " } \n",
  "\n",
  " .dataframe thead th {\n",
  "   text-align: right;\n",
  " } \n",
  "</style>\n",
  "<table border=\"1\" class=\"dataframe\">\n",
  " <thead>\n",
  " <tr style=\"text-align: right;\">\n",
  " <th></th>\n",
  " <th>gender</th>\n",
  " <th>SeniorCitizen</th>\n",
  " <th>Partner</th>\n",
  " <th>Dependents</th>\n",
  " <th>tenure</th>\n",
  " <th>PhoneService</th>\n",
  " <th>MultipleLines</th>\n",
  " <th>InternetService</th>\n",

```

```
"    <th>OnlineSecurity</th>\n",
"    <th>OnlineBackup</th>\n",
"    <th>DeviceProtection</th>\n",
"    <th>TechSupport</th>\n",
"    <th>StreamingTV</th>\n",
"    <th>StreamingMovies</th>\n",
"    <th>Contract</th>\n",
"    <th>PaperlessBilling</th>\n",
"    <th>PaymentMethod</th>\n",
"    <th>MonthlyCharges</th>\n",
"    <th>TotalCharges</th>\n",
"    <th>Churn</th>\n",
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"</thead>\n",
"<tbody>\n",
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"    <td>0</td>\n",
"    <td>0</td>
```

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" <td>0</td>\n",
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"   <td>0</td>\n",
"   <td>1</td>\n",
"   <td>2</td>\n",
"   <td>70.70</td>\n",
"   <td>151.65</td>\n",
"   <td>1</td>\n",
" </tr>\n",
" </tbody>\n",
"</table>\n",
```

```
"</div>\n",
"  <div class=\"colab-df-buttons\">\n",
"\n",
"  <div class=\"colab-df-container\">\n",
"    <button class=\"colab-df-convert\" onclick=\"convertToInteractive('df-7
09e34a1-db8d-4e06-81f9-17fe7bfb47cf')\" \n",
"      title=\"Convert this dataframe to an interactive table.\" \n",
"      style=\"display:none;\">\n",
"\n",
"    <svg xmlns=\"http://www.w3.org/2000/svg\" height=\"24px\" viewBox=
\"0 960 960 960\">\n",
"      <path d=\"M120-120v-720h720v720H120Zm60-500h600v-160H180v1
60Zm220 220h160v-160H400v160Zm0 220h160v-160H400v160ZM180-400h1
60v-160H180v160Zm440 0h160v-160H620v160ZM180-180h160v-160H180v16
0Zm440 0h160v-160H620v160Z\"/>\n",
"    </svg>\n",
"  </button>\n",
"\n",
"  <style>\n",
"    .colab-df-container {\n",
"      display: flex;\n",
"      gap: 12px;\n",
"    }\n",
"\n",
"    .colab-df-convert {\n",
"      background-color: #E8F0FE;\n",
"      border: none;\n",
"      border-radius: 50%;\n",
"      cursor: pointer;\n",
"      display: none;\n",
"      fill: #1967D2;\n",
"      height: 32px;\n",
"      padding: 0 0 0 0;\n",

```

```
" width: 32px;\n",
" }\n",
"\n",
" .colab-df-convert:hover {\n",
" background-color: #E2EBFA;\n",
" box-shadow: 0px 1px 2px rgba(60, 64, 67, 0.3), 0px 1px 3px 1px rgba(60, 64, 67, 0.15);\n",
" fill: #174EA6;\n",
" }\n",
"\n",
" .colab-df-buttons div {\n",
" margin-bottom: 4px;\n",
" }\n",
"\n",
" [theme=dark] .colab-df-convert {\n",
" background-color: #3B4455;\n",
" fill: #D2E3FC;\n",
" }\n",
"\n",
" [theme=dark] .colab-df-convert:hover {\n",
" background-color: #434B5C;\n",
" box-shadow: 0px 1px 3px 1px rgba(0, 0, 0, 0.15);\n",
" filter: drop-shadow(0px 1px 2px rgba(0, 0, 0, 0.3));\n",
" fill: #FFFFFF;\n",
" }\n",
" </style>\n",
"\n",
" <script>\n",
" const buttonEl =\n",
" document.querySelector('#df-709e34a1-db8d-4e06-81f9-17fe7bfb47cf button.colab-df-convert');\n",
" buttonEl.style.display =\n",
" google.colab.kernel.accessAllowed ? 'block' : 'none';\n",
```

```

"\n",
"  async function convertToInteractive(key) {\n",
"    const element = document.querySelector('#df-709e34a1-db8d-4e06-81f9-17fe7bfb47cf');\n",
"    const dataTable =\n",
"      await google.colab.kernel.invokeFunction('convertToInteractive',\n",
"        [key], {});\n",
"    if (!dataTable) return;\n",
"\n",
"    const docLinkHtml = 'Like what you see? Visit the ' +\n",
"      '<a target=\"_blank\" href=https://colab.research.google.com/notebooks/data_table.ipynb>data table notebook</a>'\n",
"      + ' to learn more about interactive tables.';\n",
"    element.innerHTML = \";\n",
"    dataTable['output_type'] = 'display_data';\n",
"    await google.colab.output.renderOutput(dataTable, element);\n",
"    const docLink = document.createElement('div');\n",
"    docLink.innerHTML = docLinkHtml;\n",
"    element.appendChild(docLink);\n",
"  }\n",
" </script>\n",
" </div>\n",
"\n",
"\n",
"<div id=\"df-dc38945d-1148-4e09-83b6-2c35e6dbd0ee\">\n",
"  <button class=\"colab-df-quickchart\" onclick=\"quickchart('df-dc38945d-1148-4e09-83b6-2c35e6dbd0ee')\"\n",
"    title=\"Suggest charts\"\n",
"    style=\"display:none;\">\n",
"\n",
"    <svg xmlns=\"http://www.w3.org/2000/svg\" height=\"24px\" viewBox=\"0 0 24 24\"\n",
"      width=\"24px\">\n",

```

```
" <g>\n",
"   <path d=\"M19 3H5c-1.1 0-2 .9-2 2v14c0 1.1.9 2 2 2h14c1.1 0 2-.9 2-2V
5c0-1.1-.9-2-2-2zM9 17H7v-7h2v7zm4 0h-2V7h2v10zm4 0h-2v-4h2v4z\"/>\n",
" </g>\n",
"</svg>\n",
" </button>\n",
"\n",
"<style>\n",
" .colab-df-quickchart {\n",
"   --bg-color: #E8F0FE;\n",
"   --fill-color: #1967D2;\n",
"   --hover-bg-color: #E2EBFA;\n",
"   --hover-fill-color: #174EA6;\n",
"   --disabled-fill-color: #AAA;\n",
"   --disabled-bg-color: #DDD;\n",
" }\n",
"\n",
" [theme=dark] .colab-df-quickchart {\n",
"   --bg-color: #3B4455;\n",
"   --fill-color: #D2E3FC;\n",
"   --hover-bg-color: #434B5C;\n",
"   --hover-fill-color: #FFFFFF;\n",
"   --disabled-bg-color: #3B4455;\n",
"   --disabled-fill-color: #666;\n",
" }\n",
"\n",
" .colab-df-quickchart {\n",
"   background-color: var(--bg-color);\n",
"   border: none;\n",
"   border-radius: 50%;\n",
"   cursor: pointer;\n",
"   display: none;\n",
"   fill: var(--fill-color);
```

```
" height: 32px;\n",
" padding: 0;\n",
" width: 32px;\n",
" }\n",
"\n",
" .colab-df-quickchart:hover {\n",
" background-color: var(--hover-bg-color);\n",
" box-shadow: 0 1px 2px rgba(60, 64, 67, 0.3), 0 1px 3px 1px rgba(60, 64,
67, 0.15);\n",
" fill: var(--button-hover-fill-color);\n",
" }\n",
"\n",
" .colab-df-quickchart-complete:disabled,\n",
" .colab-df-quickchart-complete:disabled:hover {\n",
" background-color: var(--disabled-bg-color);\n",
" fill: var(--disabled-fill-color);\n",
" box-shadow: none;\n",
" }\n",
"\n",
" .colab-df-spinner {\n",
" border: 2px solid var(--fill-color);\n",
" border-color: transparent;\n",
" border-bottom-color: var(--fill-color);\n",
" animation:\n",
" spin 1s steps(1) infinite;\n",
" }\n",
"\n",
" @keyframes spin {\n",
" 0% {\n",
" border-color: transparent;\n",
" border-bottom-color: var(--fill-color);\n",
" border-left-color: var(--fill-color);\n",
" }\n",
"
```

```
" 20% {\n",
"   border-color: transparent;\n",
"   border-left-color: var(--fill-color);\n",
"   border-top-color: var(--fill-color);\n",
" } \n",
" 30% {\n",
"   border-color: transparent;\n",
"   border-left-color: var(--fill-color);\n",
"   border-top-color: var(--fill-color);\n",
"   border-right-color: var(--fill-color);\n",
" } \n",
" 40% {\n",
"   border-color: transparent;\n",
"   border-right-color: var(--fill-color);\n",
"   border-top-color: var(--fill-color);\n",
" } \n",
" 60% {\n",
"   border-color: transparent;\n",
"   border-right-color: var(--fill-color);\n",
" } \n",
" 80% {\n",
"   border-color: transparent;\n",
"   border-right-color: var(--fill-color);\n",
"   border-bottom-color: var(--fill-color);\n",
" } \n",
" 90% {\n",
"   border-color: transparent;\n",
"   border-bottom-color: var(--fill-color);\n",
" } \n",
" } \n",
"</style>\n",
"\n",
" <script>\n",
```



```

" async function quickchart(key) {\n",
"   const quickchartButtonEl =\n",
"     document.querySelector('#' + key + ' button');\n",
"   quickchartButtonEl.disabled = true; // To prevent multiple clicks.\n",
"   quickchartButtonEl.classList.add('colab-df-spinner');\n",
"   try {\n",
"     const charts = await google.colab.kernel.invokeFunction(\n",
"       'suggestCharts', [key], {});\n",
"   } catch (error) {\n",
"     console.error('Error during call to suggestCharts:', error);\n",
"   }\n",
"   quickchartButtonEl.classList.remove('colab-df-spinner');\n",
"   quickchartButtonEl.classList.add('colab-df-quickchart-complete');\n",
" }\n",
" (() => {\n",
"   let quickchartButtonEl =\n",
"     document.querySelector('#df-dc38945d-1148-4e09-83b6-2c35e6d\n",
bd0ee button');\n",
"   quickchartButtonEl.style.display =\n",
"     google.colab.kernel.accessAllowed ? 'block' : 'none';\n",
"   })();\n",
" </script>\n",
"</div>\n",
"\n",
" </div>\n",
" </div>\n",
],
"application/vnd.google.colaboratory.intrinsic+json": {
  "type": "dataframe",
  "variable_name": "df",
  "summary": "{\n  \"name\": \"df\",\n  \"rows\": 7043,\n  \"fields\": [\n    {\n      \"column\": \"gender\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 0,\n        \"min\": 0,\n        \"max\": 1,\n        \"num_unique_values\": 2,\n        \"samples\"

```

```
": [\n    1,\n    0\n  ],\n  \"semantic_type\": \"\",\n  \"description\": \"\"\n}\n},\n{\n  \"column\": \"SeniorCitizen\",\n  \"properties\": {\n    \"dtype\":\n: \"number\",\n    \"std\": 0,\n    \"min\": 0,\n    \"max\": 1,\n    \"num_unique_v\nalues\": 2,\n    \"samples\": [\n    1,\n    0\n  ],\n  \"semantic_type\": \"\",\n  \"description\": \"\"\n}\n},\n{\n  \"column\": \"Partner\",\n  \"properti\nes\": {\n    \"dtype\": \"number\",\n    \"std\": 0,\n    \"min\": 0,\n    \"max\": 1,\n    \"num_unique_values\": 2,\n    \"samples\": [\n    0,\n    1\n  ],\n  \"se\nmantic_type\": \"\",\n  \"description\": \"\"\n}\n},\n{\n  \"column\": \"Dep\nendents\",\n  \"properties\": {\n    \"dtype\": \"number\",\n    \"std\": 0,\n    \"m\nin\": 0,\n    \"max\": 1,\n    \"num_unique_values\": 2,\n    \"samples\": [\n    1,\n    0\n  ],\n  \"semantic_type\": \"\",\n  \"description\": \"\"\n}\n},\n{\n  \"column\": \"tenure\",\n  \"properties\": {\n    \"dtype\": \"number\",\n    \"std\": 24,\n    \"min\": 0,\n    \"max\": 72,\n    \"num_unique_values\": 73,\n    \"samples\": [\n    8,\n    40\n  ],\n  \"semantic_type\": \"\",\n  \"descri\nption\": \"\"\n}\n},\n{\n  \"column\": \"PhoneService\",\n  \"properties\": {\n    \"dtype\": \"number\",\n    \"std\": 0,\n    \"min\": 0,\n    \"max\": 1,\n    \"n\num_unique_values\": 2,\n    \"samples\": [\n    1,\n    0\n  ],\n  \"semantic_type\": \"\",\n  \"description\": \"\"\n}\n},\n{\n  \"colu\nmn\": \"MultipleLines\",\n  \"properties\": {\n    \"dtype\": \"number\",\n    \"std\": 0,\n    \"min\": 0,\n    \"max\": 2,\n    \"num_unique_values\": 3,\n    \"sample\ns\": [\n    1,\n    0\n  ],\n  \"semantic_type\": \"\",\n  \"description\": \"\"\n}\n},\n{\n  \"column\": \"InternetService\",\n  \"properties\": {\n    \"dtype\": \"number\",\n    \"std\": 0,\n    \"min\": 0,\n    \"max\": 2,\n    \"num_uniqu\n_e_values\": 3,\n    \"samples\": [\n    0,\n    1\n  ],\n  \"semantic_type\": \"\n\",\n  \"description\": \"\"\n}\n},\n{\n  \"column\": \"OnlineSecurity\",\n  \"\nproperties\": {\n    \"dtype\": \"number\",\n    \"std\": 0,\n    \"min\": 0,\n    \"\nmax\": 2,\n    \"num_unique_values\": 3,\n    \"samples\": [\n    0,\n    2\n  ],\n  \"semantic_type\": \"\",\n  \"description\": \"\"\n}\n},\n{\n  \"colu\nmn\": \"OnlineBackup\",\n  \"properties\": {\n    \"dtype\": \"number\",\n    \"std\": 0,\n    \"min\": 0,\n    \"max\": 2,\n    \"num_unique_values\": 3,\n    \"sample\ns\": [\n    2,\n    0\n  ],\n  \"semantic_type\": \"\",\n  \"description\": \"\"\n}\n},\n{\n  \"column\": \"DeviceProtection\",\n  \"properties\": {\n    \"\ndtype\": \"number\",\n    \"std\": 0,\n    \"min\": 0,\n    \"max\": 2,\n    \"num_un
```

```
ique_values": 3,\n  "samples": [\n    0,\n    2\n  ],\n  "semantic_type": "\n  {\n    \"description\": \"TechSupport\",,\n    \"properties\": {\n      \"dtype\": \"number\",,\n      \"std\": 0,\n      \"min\": 0,\n      \"max\": 2,\n      \"num_unique_values\": 3,\n      \"samples\": [\n        0,\n        2\n      ],,\n      \"semantic_type\": \"\n    },,\n    {\n      \"column\": \"StreamingTV\",,\n      \"properties\": {\n        \"dtype\": \"number\",,\n        \"std\": 0,\n        \"min\": 0,\n        \"max\": 2,\n        \"num_unique_values\": 3,\n        \"samples\": [\n          0,\n          2\n        ],,\n        \"semantic_type\": \"\n      },,\n      {\n        \"column\": \"StreamingMovies\",,\n        \"properties\": {\n          \"dtype\": \"number\",,\n          \"std\": 0,\n          \"min\": 0,\n          \"max\": 2,\n          \"num_unique_values\": 3,\n          \"samples\": [\n            0,\n            2\n          ],,\n          \"semantic_type\": \"\n        },,\n        {\n          \"column\": \"Contract\",,\n          \"properties\": {\n            \"dtype\": \"number\",,\n            \"std\": 0,\n            \"min\": 0,\n            \"max\": 2,\n            \"num_unique_values\": 3,\n            \"samples\": [\n              0,\n              1\n            ],,\n            \"semantic_type\": \"\n          },,\n          {\n            \"column\": \"PaperlessBilling\",,\n            \"properties\": {\n              \"dtype\": \"number\",,\n              \"std\": 0,\n              \"min\": 0,\n              \"max\": 1,\n              \"num_unique_values\": 2,\n              \"samples\": [\n                0,\n                1\n              ],,\n              \"semantic_type\": \"\n            },,\n            {\n              \"column\": \"PaymentMethod\",,\n              \"properties\": {\n                \"dtype\": \"number\",,\n                \"std\": 1,\n                \"min\": 0,\n                \"max\": 3,\n                \"num_unique_values\": 4,\n                \"samples\": [\n                  3,\n                  1\n                ],,\n                \"semantic_type\": \"\n              },,\n              {\n                \"column\": \"MonthlyCharges\",,\n                \"properties\": {\n                  \"dtype\": \"number\",,\n                  \"std\": 30.09004709767854,\n                  \"min\": 18.25,\n                  \"max\": 118.75,\n                  \"num_unique_-\n                },,\n                {\n                  \"column\": \"TotalCharges\",,\n                  \"properties\": {\n                    \"dtype\": \"number\",,\n                    \"std\": 2266.7944696890195,\n                    \"min\": 0.0,\n                    \"max\": 8684.8,\n                    \"num_unique_values\": 6531,\n                    \"samples\": [\n                      4600.7,\n                      20.35\n                    ],,\n                    \"semantic_type\": \"\n                  },,\n                  {\n                    \"column\": \"Churn\",,\n                    \"properties\": {\n                      \"dtype\": \"number\",,\n                      \"std\": 0,\n                      \"min\": 0,\n                      \"max\": 1,\n                      \"num_unique_values\": 2,\n                      \"samples\": [\n                        1,\n                        0\n                      ],,\n
```

```

    "semantic_type": "\n",
    "description": "\n  }\n  }\n ]\n}"
  },
  "metadata": {},
  "execution_count": 64
}
],
},
{
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  "source": [
    "**Traianing and test data split**"
  ],
  "metadata": {
    "id": "hEURc8NrbufR"
  },
},
{
  "cell_type": "code",
  "source": [
    "# splitting the features and target\n",
    "X = df.drop(columns=[\"Churn\"])\n",
    "y = df[\"Churn\"]"
  ],
  "metadata": {
    "id": "wN0ylxzKaUyE"
  },
},
"execution_count": 65,
"outputs": []
},
{
  "cell_type": "code",
  "source": [

```

```
"# split training and test data\n",  
"X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_  
state=42)"  
],  
"metadata": {  
  "id": "buwUAYVUb4aR"  
},  
"execution_count": 69,  
"outputs": []  
},  
{  
  "cell_type": "code",  
  "source": [  
    "print(y_train.shape)"  
  ],  
  "metadata": {  
    "colab": {  
      "base_uri": "https://localhost:8080/"  
    },  
    "id": "vp5UEF3PcObc",  
    "outputId": "28bb4813-f6f8-4d38-e122-0bee4afd73f1"  
  },  
  "execution_count": 70,  
  "outputs": [  
    {  
      "output_type": "stream",  
      "name": "stdout",  
      "text": [  
        "(5634,)\n"  
      ]  
    }  
  ]  
},
```

```
{
  "cell_type": "code",
  "source": [
    "print(y_train.value_counts())"
  ],
  "metadata": {
    "colab": {
      "base_uri": "https://localhost:8080/"
    },
    "id": "GzXErq7jcbaJ",
    "outputId": "ab99c3f8-62eb-4f73-90c7-ff6fd26aec5c"
  },
  "execution_count": 71,
  "outputs": [
    {
      "output_type": "stream",
      "name": "stdout",
      "text": [
        "Churn\n",
        "0    4138\n",
        "1    1496\n",
        "Name: count, dtype: int64\n"
      ]
    }
  ],
},
{
  "cell_type": "markdown",
  "source": [
    "Synthetic Minority Oversampling TEchnique (SMOTE)"
  ],
  "metadata": {
    "id": "SW6dcHZMckA6"
  }
}
```

```
}  
,  
{  
  "cell_type": "code",  
  "source": [  
    "smote = SMOTE(random_state=42)"  
  ],  
  "metadata": {  
    "id": "T7rj92Opcgh_"  
  },  
  "execution_count": 72,  
  "outputs": []  
},  
{  
  "cell_type": "code",  
  "source": [  
    "X_train_smote, y_train_smote = smote.fit_resample(X_train, y_train)"  
  ],  
  "metadata": {  
    "id": "oZkUnYCEcxLL"  
  },  
  "execution_count": 73,  
  "outputs": []  
},  
{  
  "cell_type": "code",  
  "source": [  
    "print(y_train_smote.shape)"  
  ],  
  "metadata": {  
    "colab": {  
      "base_uri": "https://localhost:8080/"  
    },  
  },  
}
```

```
"id": "QHiRrR3Jc9Uv",
"outputId": "43a76b42-f7e8-41fe-b94c-950ecc64de62"
},
"execution_count": 74,
"outputs": [
  {
    "output_type": "stream",
    "name": "stdout",
    "text": [
      "(8276,)\n"
    ]
  }
],
},
{
  "cell_type": "code",
  "source": [
    "print(y_train_smote.value_counts())"
  ],
  "metadata": {
    "colab": {
      "base_uri": "https://localhost:8080/"
    },
    "id": "IMS-DWO0dA17",
    "outputId": "578b3563-742e-436f-9158-2c3217ed1840"
  },
  "execution_count": 75,
  "outputs": [
    {
      "output_type": "stream",
      "name": "stdout",
      "text": [
        "Churn\n",

```



```
"0  4138\n",
"1  4138\n",
"Name: count, dtype: int64\n"
]
}
],
{
"cell_type": "markdown",
"source": [
  "***5. Model Training**"
],
"metadata": {
  "id": "paChOpy7dNV9"
}
},
{
"cell_type": "markdown",
"source": [
  "Training with default hyperparameters"
],
"metadata": {
  "id": "9XoFtHYxdW3n"
}
},
{
"cell_type": "code",
"source": [
  "# dictionary of models\n",
  "models = {\n",
  "  \"Decision Tree\": DecisionTreeClassifier(random_state=42),\n",
  "  \"Random Forest\": RandomForestClassifier(random_state=42),\n",
  "  \"XGBoost\": XGBClassifier(random_state=42)\n",
}
```

```

    "}"
],
"metadata": {
    "id": "6zy42grTdEQI"
},
"execution_count": 76,
"outputs": []
},
{
    "cell_type": "code",
    "source": [
        "# dictionary to store the cross validation results\n",
        "cv_scores = {}\n",
        "\n",
        "# perform 5-fold cross validation for each model\n",
        "for model_name, model in models.items():\n",
        "    print(f"Training {model_name} with default parameters")\n",
        "    scores = cross_val_score(model, X_train_smote, y_train_smote, cv=5, scor
ing="accuracy")\n",
        "    cv_scores[model_name] = scores\n",
        "    print(f"{model_name} cross-validation accuracy: {np.mean(scores):.2f}")
\n",
        "    print(\"-\"*70)"
    ],
    "metadata": {
        "colab": {
            "base_uri": "https://localhost:8080/"
        },
        "id": "B3KGjoiWdkUY",
        "outputId": "9bd95a2a-51bc-4c86-dee4-cd7f8597ccfb"
    },
    "execution_count": 81,
    "outputs": [

```

```

{
  "output_type": "stream",
  "name": "stdout",
  "text": [
    "Training Decision Tree with default parameters\n",
    "Decision Tree cross-validation accuracy: 0.78\n",
    "-----\n",
    "Training Random Forest with default parameters\n",
    "Random Forest cross-validation accuracy: 0.84\n",
    "-----\n",
    "Training XGBoost with default parameters\n",
    "XGBoost cross-validation accuracy: 0.83\n",
    "-----\n"
  ]
}

],
{
  "cell_type": "code",
  "source": [
    "cv_scores"
  ],
  "metadata": {
    "colab": {
      "base_uri": "https://localhost:8080/"
    },
    "id": "azUmCrtsd7X9",
    "outputId": "d74e6478-85a5-457b-a303-65d8c3ec1de7"
  },
  "execution_count": 82,
  "outputs": [
    {
      "output_type": "execute_result",

```

```
"data": {
  "text/plain": [
    '{Decision Tree': array([0.68297101, 0.71299094, 0.82175227, 0.8356495
5, 0.83564955]),\n",
    "'Random Forest': array([0.72524155, 0.77824773, 0.90513595, 0.89425
982, 0.90090634]),\n",
    "'XGBoost': array([0.70048309, 0.75649547, 0.90271903, 0.89486405,
0.90030211])}"
  ]
},
"metadata": {},
"execution_count": 82
}
],
{
  "cell_type": "markdown",
  "source": [
    "Random Forest gives the highest accuracy compared to other models with d
efault parameters"
  ],
  "metadata": {
    "id": "GZHMA7qDfdhy"
  }
},
{
  "cell_type": "code",
  "source": [
    "rfc = RandomForestClassifier(random_state=42)"
  ],
  "metadata": {
    "id": "7Ryv2eVge7jo"
  }
},
```

```
"execution_count": 88,
"outputs": []
},
{
  "cell_type": "code",
  "source": [
    "rfc.fit(X_train_smote, y_train_smote)"
  ],
  "metadata": {
    "colab": {
      "base_uri": "https://localhost:8080/",
      "height": 80
    },
    "id": "LtSf0RDCfoGj",
    "outputId": "315c8fa4-9e37-47f7-ce47-4c93d6c52c2a"
  },
  "execution_count": 89,
  "outputs": [
    {
      "output_type": "execute_result",
      "data": {
        "text/plain": [
          "RandomForestClassifier(random_state=42)"
        ],
        "text/html": [
          "<style>#sk-container-id-2 {\n",
          " /* Definition of color scheme common for light and dark mode */\n",
          " --sklearn-color-text: black;\n",
          " --sklearn-color-line: gray;\n",
          " /* Definition of color scheme for unfitted estimators */\n",
          " --sklearn-color-unfitted-level-0: #fff5e6;\n",
          " --sklearn-color-unfitted-level-1: #f6e4d2;\n",
          " --sklearn-color-unfitted-level-2: #ffe0b3;\n",

```

```

" --sklearn-color-unfitted-level-3: chocolate;\n",
" /* Definition of color scheme for fitted estimators */\n",
" --sklearn-color-fitted-level-0: #f0f8ff;\n",
" --sklearn-color-fitted-level-1: #d4ebff;\n",
" --sklearn-color-fitted-level-2: #b3dbfd;\n",
" --sklearn-color-fitted-level-3: cornflowerblue;\n",
"\n",
" /* Specific color for light theme */\n",
" --sklearn-color-text-on-default-background: var(--sg-text-color, var(--t
heme-code-foreground, var(--jp-content-font-color1, black))); \n",
" --sklearn-color-background: var(--sg-background-color, var(--theme-b
ackground, var(--jp-layout-color0, white))); \n",
" --sklearn-color-border-box: var(--sg-text-color, var(--theme-code-fore
ground, var(--jp-content-font-color1, black))); \n",
" --sklearn-color-icon: #696969;\n",
"\n",
" @media (prefers-color-scheme: dark) {\n",
" /* Redefinition of color scheme for dark theme */\n",
" --sklearn-color-text-on-default-background: var(--sg-text-color, var(--t
heme-code-foreground, var(--jp-content-font-color1, white))); \n",
" --sklearn-color-background: var(--sg-background-color, var(--theme-
background, var(--jp-layout-color0, #111))); \n",
" --sklearn-color-border-box: var(--sg-text-color, var(--theme-code-fore
ground, var(--jp-content-font-color1, white))); \n",
" --sklearn-color-icon: #878787;\n",
" }\n",
"}\n",
"\n",
"#sk-container-id-2 {\n",
" color: var(--sklearn-color-text);\n",
"}\n",
"\n",
"#sk-container-id-2 pre {\n",

```

```
" padding: 0;\n",
"}\n",
"\n",
"#sk-container-id-2 input.sk-hidden--visually {\n",
" border: 0;\n",
" clip: rect(1px 1px 1px 1px);\n",
" clip: rect(1px, 1px, 1px, 1px);\n",
" height: 1px;\n",
" margin: -1px;\n",
" overflow: hidden;\n",
" padding: 0;\n",
" position: absolute;\n",
" width: 1px;\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-dashed-wrapped {\n",
" border: 1px dashed var(--sklearn-color-line);\n",
" margin: 0 0.4em 0.5em 0.4em;\n",
" box-sizing: border-box;\n",
" padding-bottom: 0.4em;\n",
" background-color: var(--sklearn-color-background);\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-container {\n",
" /* jupyter's `normalize.less` sets `[hidden] { display: none; }`\n",
" but bootstrap.min.css set `[hidden] { display: none !important; }`\n",
" so we also need the `!important` here to be able to override the\n",
" default hidden behavior on the sphinx rendered scikit-learn.org.\n",
" See: https://github.com/scikit-learn/scikit-learn/issues/21755 */\n",
" display: inline-block !important;\n",
" position: relative;\n",
"}\n",
"\n",
```

```

"#sk-container-id-2 div.sk-text-repr-fallback {\n",
" display: none;\n",
"}\n",
"\n",
"div.sk-parallel-item,\n",
"div.sk-serial,\n",
"div.sk-item {\n",
" /* draw centered vertical line to link estimators */\n",
" background-image: linear-gradient(var(--sklearn-color-text-on-default-
background), var(--sklearn-color-text-on-default-background));\n",
" background-size: 2px 100%;\n",
" background-repeat: no-repeat;\n",
" background-position: center center;\n",
"}\n",
"\n",
"/* Parallel-specific style estimator block */\n",
"\n",
"#sk-container-id-2 div.sk-parallel-item::after {\n",
" content: \"\";\n",
" width: 100%;\n",
" border-bottom: 2px solid var(--sklearn-color-text-on-default-backgroun
d);\n",
" flex-grow: 1;\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-parallel {\n",
" display: flex;\n",
" align-items: stretch;\n",
" justify-content: center;\n",
" background-color: var(--sklearn-color-background);\n",
" position: relative;\n",
"}\n",
"\n",

```



```
"#sk-container-id-2 div.sk-parallel-item {\n",
"  display: flex;\n",
"  flex-direction: column;\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-parallel-item:first-child::after {\n",
"  align-self: flex-end;\n",
"  width: 50%;\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-parallel-item:last-child::after {\n",
"  align-self: flex-start;\n",
"  width: 50%;\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-parallel-item:only-child::after {\n",
"  width: 0;\n",
"}\n",
"\n",
"/* Serial-specific style estimator block */\n",
"\n",
"#sk-container-id-2 div.sk-serial {\n",
"  display: flex;\n",
"  flex-direction: column;\n",
"  align-items: center;\n",
"  background-color: var(--sklearn-color-background);\n",
"  padding-right: 1em;\n",
"  padding-left: 1em;\n",
"}\n",
"\n",
"\n",
"/* Toggleable style: style used for estimator/Pipeline/ColumnTransformer  
box that is\n",
```

```

"clickable and can be expanded/collapsed.\n",
"- Pipeline and ColumnTransformer use this feature and define the default style\n",
"- Estimators will overwrite some part of the style using the `sk-estimator` class\n",
"*/\n",
"\n",
"/* Pipeline and ColumnTransformer style (default) */\n",
"\n",
"#sk-container-id-2 div.sk-toggleable {\n",
" /* Default theme specific background. It is overwritten whether we have a\n",
"\n",
"  specific estimator or a Pipeline/ColumnTransformer */\n",
"  background-color: var(--sklearn-color-background);\n",
"}\n",
"\n",
"/* Toggleable label */\n",
"#sk-container-id-2 label.sk-toggleable__label {\n",
"  cursor: pointer;\n",
"  display: block;\n",
"  width: 100%;\n",
"  margin-bottom: 0;\n",
"  padding: 0.5em;\n",
"  box-sizing: border-box;\n",
"  text-align: center;\n",
"}\n",
"\n",
"#sk-container-id-2 label.sk-toggleable__label-arrow:before {\n",
" /* Arrow on the left of the label */\n",
"  content: \">\";\n",
"  float: left;\n",
"  margin-right: 0.25em;\n",
"  color: var(--sklearn-color-icon);\n",

```

```
"}\n",
"\n",
"#sk-container-id-2 label.sk-toggleable__label-arrow:hover:before {\n",
"  color: var(--sklearn-color-text);\n",
"}\n",
"\n",
"/* Toggleable content - dropdown */\n",
"\n",
"#sk-container-id-2 div.sk-toggleable__content {\n",
"  max-height: 0;\n",
"  max-width: 0;\n",
"  overflow: hidden;\n",
"  text-align: left;\n",
"  /* unfitted */\n",
"  background-color: var(--sklearn-color-unfitted-level-0);\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-toggleable__content.fitted {\n",
"  /* fitted */\n",
"  background-color: var(--sklearn-color-fitted-level-0);\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-toggleable__content pre {\n",
"  margin: 0.2em;\n",
"  border-radius: 0.25em;\n",
"  color: var(--sklearn-color-text);\n",
"  /* unfitted */\n",
"  background-color: var(--sklearn-color-unfitted-level-0);\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-toggleable__content.fitted pre {\n",
"  /* unfitted */\n",
"  background-color: var(--sklearn-color-fitted-level-0);\n",
"}
```

```

    "}\n",
    "\n",
    "#sk-container-id-2 input.sk-toggleable__control:checked~div.sk-toggleable__content {\n",
    " /* Expand drop-down */\n",
    " max-height: 200px;\n",
    " max-width: 100%;\n",
    " overflow: auto;\n",
    "}\n",
    "\n",
    "#sk-container-id-2 input.sk-toggleable__control:checked~label.sk-toggleable__label-arrow:before {\n",
    " content: \"▼\";\n",
    "}\n",
    "\n",
    "/* Pipeline/ColumnTransformer-specific style */\n",
    "\n",
    "#sk-container-id-2 div.sk-label input.sk-toggleable__control:checked~label.sk-toggleable__label {\n",
    " color: var(--sklearn-color-text);\n",
    " background-color: var(--sklearn-color-unfitted-level-2);\n",
    "}\n",
    "\n",
    "#sk-container-id-2 div.sk-label.fitted input.sk-toggleable__control:checked~label.sk-toggleable__label {\n",
    " background-color: var(--sklearn-color-fitted-level-2);\n",
    "}\n",
    "\n",
    "/* Estimator-specific style */\n",
    "\n",
    "/* Colorize estimator box */\n",
    "#sk-container-id-2 div.sk-estimator input.sk-toggleable__control:checked~label.sk-toggleable__label {\n",

```

```

" /* unfitted */\n",
" background-color: var(--sklearn-color-unfitted-level-2);\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-estimator.fitted input.sk-toggleable__control:ch
ecked~label.sk-toggleable__label {\n",
" /* fitted */\n",
" background-color: var(--sklearn-color-fitted-level-2);\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-label label.sk-toggleable__label,\n",
"#sk-container-id-2 div.sk-label label {\n",
" /* The background is the default theme color */\n",
" color: var(--sklearn-color-text-on-default-background);\n",
"}\n",
"\n",
"/* On hover, darken the color of the background */\n",
"#sk-container-id-2 div.sk-label:hover label.sk-toggleable__label {\n",
" color: var(--sklearn-color-text);\n",
" background-color: var(--sklearn-color-unfitted-level-2);\n",
"}\n",
"\n",
"/* Label box, darken color on hover, fitted */\n",
"#sk-container-id-2 div.sk-label.fitted:hover label.sk-toggleable__label.fitt
ed {\n",
" color: var(--sklearn-color-text);\n",
" background-color: var(--sklearn-color-fitted-level-2);\n",
"}\n",
"\n",
"/* Estimator label */\n",
"\n",
"#sk-container-id-2 div.sk-label label {\n",
" font-family: monospace;\n",

```

```
" font-weight: bold;\n",
" display: inline-block;\n",
" line-height: 1.2em;\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-label-container {\n",
" text-align: center;\n",
"}\n",
"\n",
"/* Estimator-specific */\n",
"#sk-container-id-2 div.sk-estimator {\n",
" font-family: monospace;\n",
" border: 1px dotted var(--sklearn-color-border-box);\n",
" border-radius: 0.25em;\n",
" box-sizing: border-box;\n",
" margin-bottom: 0.5em;\n",
" /* unfitted */\n",
" background-color: var(--sklearn-color-unfitted-level-0);\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-estimator.fitted {\n",
" /* fitted */\n",
" background-color: var(--sklearn-color-fitted-level-0);\n",
"}\n",
"\n",
"/* on hover */\n",
"#sk-container-id-2 div.sk-estimator:hover {\n",
" /* unfitted */\n",
" background-color: var(--sklearn-color-unfitted-level-2);\n",
"}\n",
"\n",
"#sk-container-id-2 div.sk-estimator.fitted:hover {\n",
" /* fitted */\n",
```

```
" background-color: var(--sklearn-color-fitted-level-2);\n",\n"}\n",\n"\n",\n"/\n/* Specification for estimator info (e.g. \"i\" and \"?\") */\n",\n"\n",\n"/\n/* Common style for \"i\" and \"?\" */\n",\n"\n",\n".sk-estimator-doc-link,\n",\n"a:link.sk-estimator-doc-link,\n",\n"a:visited.sk-estimator-doc-link {\n",\n" float: right;\n",\n" font-size: smaller;\n",\n" line-height: 1em;\n",\n" font-family: monospace;\n",\n" background-color: var(--sklearn-color-background);\n",\n" border-radius: 1em;\n",\n" height: 1em;\n",\n" width: 1em;\n",\n" text-decoration: none !important;\n",\n" margin-left: 1ex;\n",\n" /\n/* unfitted */\n",\n" border: var(--sklearn-color-unfitted-level-1) 1pt solid;\n",\n" color: var(--sklearn-color-unfitted-level-1);\n",\n"}\n",\n"\n",\n".sk-estimator-doc-link.fitted,\n",\n"a:link.sk-estimator-doc-link.fitted,\n",\n"a:visited.sk-estimator-doc-link.fitted {\n",\n" /\n/* fitted */\n",\n" border: var(--sklearn-color-fitted-level-1) 1pt solid;\n",\n" color: var(--sklearn-color-fitted-level-1);\n",\n"}\n",\n"\n",
```

```
"/* On hover */\n",
"div.sk-estimator: hover .sk-estimator-doc-link: hover,\n",
".sk-estimator-doc-link: hover,\n",
"div.sk-label-container: hover .sk-estimator-doc-link: hover,\n",
".sk-estimator-doc-link: hover {\n",
" /* unfitted */\n",
" background-color: var(--sklearn-color-unfitted-level-3);\n",
" color: var(--sklearn-color-background);\n",
" text-decoration: none;\n",
"}\n",
"\n",
"div.sk-estimator.fitted: hover .sk-estimator-doc-link.fitted: hover,\n",
".sk-estimator-doc-link.fitted: hover,\n",
"div.sk-label-container: hover .sk-estimator-doc-link.fitted: hover,\n",
".sk-estimator-doc-link.fitted: hover {\n",
" /* fitted */\n",
" background-color: var(--sklearn-color-fitted-level-3);\n",
" color: var(--sklearn-color-background);\n",
" text-decoration: none;\n",
"}\n",
"\n",
"/* Span, style for the box shown on hovering the info icon */\n",
".sk-estimator-doc-link span {\n",
" display: none;\n",
" z-index: 9999;\n",
" position: relative;\n",
" font-weight: normal;\n",
" right: .2ex;\n",
" padding: .5ex;\n",
" margin: .5ex;\n",
" width: min-content;\n",
" min-width: 20ex;\n",
" max-width: 50ex;\n",
}
```



```
" color: var(--sklearn-color-text);\n",\n" box-shadow: 2pt 2pt 4pt #999;\n",\n" /* unfitted */\n",\n" background: var(--sklearn-color-unfitted-level-0);\n",\n" border: .5pt solid var(--sklearn-color-unfitted-level-3);\n",\n"}\n",\n"\n",\n".sk-estimator-doc-link.fitted span {\n",\n" /* fitted */\n",\n" background: var(--sklearn-color-fitted-level-0);\n",\n" border: var(--sklearn-color-fitted-level-3);\n",\n"}\n",\n"\n",\n".sk-estimator-doc-link:hover span {\n",\n" display: block;\n",\n"}\n",\n"\n",\n"/* \"?\"-specific style due to the `<a>` HTML tag */\n",\n"\n",\n"#sk-container-id-2 a.estimator_doc_link {\n",\n" float: right;\n",\n" font-size: 1rem;\n",\n" line-height: 1em;\n",\n" font-family: monospace;\n",\n" background-color: var(--sklearn-color-background);\n",\n" border-radius: 1rem;\n",\n" height: 1rem;\n",\n" width: 1rem;\n",\n" text-decoration: none;\n",\n" /* unfitted */\n",\n" color: var(--sklearn-color-unfitted-level-1);\n",\n" border: var(--sklearn-color-unfitted-level-1) 1pt solid;\n",\n"}\n",
```

```

"\n",
"#sk-container-id-2 a.estimator_doc_link.fitted {\n",
" /* fitted */\n",
" border: var(--sklearn-color-fitted-level-1) 1pt solid;\n",
" color: var(--sklearn-color-fitted-level-1);\n",
"}\n",
"\n",
"/* On hover */\n",
"#sk-container-id-2 a.estimator_doc_link:hover {\n",
" /* unfitted */\n",
" background-color: var(--sklearn-color-unfitted-level-3);\n",
" color: var(--sklearn-color-background);\n",
" text-decoration: none;\n",
"}\n",
"\n",
"#sk-container-id-2 a.estimator_doc_link.fitted:hover {\n",
" /* fitted */\n",
" background-color: var(--sklearn-color-fitted-level-3);\n",
"}\n",

```

```

"</style><div id=\"sk-container-id-2\" class=\"sk-top-container\"><div clas
s=\"sk-text-repr-fallback\"><pre>RandomForestClassifier(random_state=42)</p
re><b>In a Jupyter environment, please rerun this cell to show the HTML represen
tation or trust the notebook. <br />On GitHub, the HTML representation is unable
to render, please try loading this page with nbviewer.org.</b></div><div class=\"
sk-container\" hidden><div class=\"sk-item\"><div class=\"sk-estimator fitted sk-
toggleable\"><input class=\"sk-toggleable__control sk-hidden--visually\" id=\"sk
-estimator-id-2\" type=\"checkbox\" checked><label for=\"sk-estimator-id-2\" cla
ss=\"sk-toggleable__label fitted sk-toggleable__label-arrow fitted\">&nbsp;&nb
sp;RandomForestClassifier<a class=\"sk-estimator-doc-link fitted\" rel=\"noreferrer
er\" target=\"_blank\" href=\"https://scikit-learn.org/1.5/modules/generated/sklea
rn.ensemble.RandomForestClassifier.html\">?<span>Documentation for Rando
mForestClassifier</span></a><span class=\"sk-estimator-doc-link fitted\">i<sp
an>Fitted</span></span></label><div class=\"sk-toggleable__content fitted\">

```

```
<pre>RandomForestClassifier(random_state=42)</pre></div> </div></div></div>
iv></div>"
    ]
    },
    "metadata": {},
    "execution_count": 89
  }
]
},
{
  "cell_type": "code",
  "source": [
    "print(y_test.value_counts())"
  ],
  "metadata": {
    "colab": {
      "base_uri": "https://localhost:8080/"
    },
    "id": "Ps-zl6NFgEJg",
    "outputId": "9ece5fd7-2825-45de-ff6b-9fc6b4ec5c35"
  },
  "execution_count": 90,
  "outputs": [
    {
      "output_type": "stream",
      "name": "stdout",
      "text": [
        "Churn\n",
        "0    1036\n",
        "1     373\n",
        "Name: count, dtype: int64\n"
      ]
    }
  ]
}
```

```

    ]
  },
  {
    "cell_type": "markdown",
    "source": [
      "**6. Model Evaluation**"
    ],
    "metadata": {
      "id": "GjsS6hgkfru0"
    }
  },
  {
    "cell_type": "code",
    "source": [
      "# evaluate on test data\n",
      "y_test_pred = rfc.predict(X_test)\n",
      "\n",
      "print(\"Accuracy Score:\\n\", accuracy_score(y_test, y_test_pred))\n",
      "print(\"Confusion Matrix:\\n\", confusion_matrix(y_test, y_test_pred))\n",
      "print(\"Classification Report:\\n\", classification_report(y_test, y_test_pred))
    ],
    "metadata": {
      "colab": {
        "base_uri": "https://localhost:8080/"
      },
      "id": "yQtGX6PngNJm",
      "outputId": "ad8ffbe2-514c-48b5-e6dd-ea9b9288a065"
    },
    "execution_count": 91,
    "outputs": [
      {
        "output_type": "stream",

```

```

"name": "stdout",
"text": [
  "Accuracy Score:\n",
  " 0.7785663591199432\n",
  "Confusion Matrix:\n",
  " [[878 158]\n",
  " [154 219]]\n",
  "Classification Report:\n",
  "      precision  recall f1-score  support\n",
  "\n",
  "    0    0.85    0.85    0.85    1036\n",
  "    1    0.58    0.59    0.58     373\n",
  "\n",
  " accuracy                0.78    1409\n",
  " macro avg    0.72    0.72    0.72    1409\n",
  "weighted avg    0.78    0.78    0.78    1409\n",
  "\n"
]
}
],
{
  "cell_type": "code",
  "source": [
    "# save the trained model as a pickle file\n",
    "model_data = {\n\"model\": rfc, \n\"features_names\": X.columns.tolist()\n}\n",
    "\n",
    "\n",
    "with open(\"customer_churn_model.pkl\", \"wb\") as f:\n",
    "    pickle.dump(model_data, f)"
  ],
  "metadata": {
    "id": "zssUNtcsgMLh"
  }
}

```

```
},
"execution_count": 94,
"outputs": []
},
{
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    "**7. Load the saved model and build a Predictive System**"
  ],
  "metadata": {
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  }
},
{
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  "source": [
    "# load teh saved model and the feature names\n",
    "\n",
    "with open(\"customer_churn_model.pkl\", \"rb\") as f:\n",
    "    model_data = pickle.load(f)\n",
    "\n",
    "loaded_model = model_data[\"model\"]\n",
    "feature_names = model_data[\"features_names\"]"
  ],
  "metadata": {
    "id": "EHJUOCxAg_-W"
  },
  "execution_count": 99,
  "outputs": []
},
{
  "cell_type": "code",
  "source": [
```

```
"print(loaded_model)"
],
"metadata": {
  "colab": {
    "base_uri": "https://localhost:8080/"
  },
  "id": "UkaI2oE3huq5",
  "outputId": "343386c6-aad8-4eac-a40d-9a38ef54023d"
},
"execution_count": 100,
"outputs": [
  {
    "output_type": "stream",
    "name": "stdout",
    "text": [
      "RandomForestClassifier(random_state=42)\n"
    ]
  }
],
},
{
  "cell_type": "code",
  "source": [
    "print(feature_names)"
  ],
  "metadata": {
    "colab": {
      "base_uri": "https://localhost:8080/"
    },
    "id": "a758RINVh7Xe",
    "outputId": "1456a90a-2599-49ec-e894-7627d3f0f919"
  },
  "execution_count": 101,
```

```
"outputs": [  
  {  
    "output_type": "stream",  
    "name": "stdout",  
    "text": [  
      ["gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure', 'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod', 'MonthlyCharges', 'TotalCharges']\n"  
    ]  
  }  
],  
{  
  "cell_type": "code",  
  "source": [  
    "input_data = {\n",  
    "  'gender': 'Female',\n",  
    "  'SeniorCitizen': 0,\n",  
    "  'Partner': 'Yes',\n",  
    "  'Dependents': 'No',\n",  
    "  'tenure': 1,\n",  
    "  'PhoneService': 'No',\n",  
    "  'MultipleLines': 'No phone service',\n",  
    "  'InternetService': 'DSL',\n",  
    "  'OnlineSecurity': 'No',\n",  
    "  'OnlineBackup': 'Yes',\n",  
    "  'DeviceProtection': 'No',\n",  
    "  'TechSupport': 'No',\n",  
    "  'StreamingTV': 'No',\n",  
    "  'StreamingMovies': 'No',\n",  
    "  'Contract': 'Month-to-month',\n",  
    "  'PaperlessBilling': 'Yes',
```



```

" 'PaymentMethod': 'Electronic check',\n",
" 'MonthlyCharges': 29.85,\n",
" 'TotalCharges': 29.85\n",
"}\n",
"\n",
"\n",
"input_data_df = pd.DataFrame([input_data])\n",
"\n",
"with open(\"encoders.pkl\", \"rb\") as f:\n",
" encoders = pickle.load(f)\n",
"\n",
"\n",
"# encode categorical features using the saved encoders\n",
"for column, encoder in encoders.items():\n",
" input_data_df[column] = encoder.transform(input_data_df[column])\n",
"\n",
"# make a prediction\n",
"prediction = loaded_model.predict(input_data_df)\n",
"pred_prob = loaded_model.predict_proba(input_data_df)\n",
"\n",
"print(prediction)\n",
"\n",
"# results\n",
"print(f\"Prediction: {'Churn' if prediction[0] == 1 else 'No Churn'}\")\n",
"print(f\"Prediction Probability: {pred_prob}\")
],
"metadata": {
"colab": {
"base_uri": "https://localhost:8080/"
},
"collapsed": true,
"id": "-zyAmZeyioWS",
"outputId": "39751cc2-d16d-466c-af3a-c0b41a3219f0"

```

```
},
"execution_count": 109,
"outputs": [
  {
    "output_type": "stream",
    "name": "stdout",
    "text": [
      "[0]\n",
      "Prediction: No Churn\n",
      "Prediciton Probability: [[0.78 0.22]]\n"
    ]
  }
],
},
{
  "cell_type": "code",
  "source": [
    "encoders"
  ],
  "metadata": {
    "colab": {
      "base_uri": "https://localhost:8080/"
    },
    "id": "pVYWRbYwjS3y",
    "outputId": "e4c606d9-4b6e-4dc9-e467-2e61314ef874"
  },
  "execution_count": 108,
  "outputs": [
    {
      "output_type": "execute_result",
      "data": {
        "text/plain": [
          "{ 'gender': LabelEncoder(),\n",
```

```

    " 'Partner': LabelEncoder(),\n",
    " 'Dependents': LabelEncoder(),\n",
    " 'PhoneService': LabelEncoder(),\n",
    " 'MultipleLines': LabelEncoder(),\n",
    " 'InternetService': LabelEncoder(),\n",
    " 'OnlineSecurity': LabelEncoder(),\n",
    " 'OnlineBackup': LabelEncoder(),\n",
    " 'DeviceProtection': LabelEncoder(),\n",
    " 'TechSupport': LabelEncoder(),\n",
    " 'StreamingTV': LabelEncoder(),\n",
    " 'StreamingMovies': LabelEncoder(),\n",
    " 'Contract': LabelEncoder(),\n",
    " 'PaperlessBilling': LabelEncoder(),\n",
    " 'PaymentMethod': LabelEncoder()}"
]
},
"metadata": {},
"execution_count": 108
}
]
},
{
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"source": [
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    "1. Implement Hyperparameter Tuining\n",
    "2. Try Model Selection\n",
    "3. Try downsampling\n",
    "4. Try to address teh overfitting\n",
    "5. Try Startified k fold CV"
],
"metadata": {
    "id": "wvQG7N02IKFV"
}

```

```
}  
,  
{  
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  },  
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  "outputs": []  
}  
]  
}
```